EMITO-METRIX APPLICATION: QUICK START TUTORIAL

EMito-Metrix is a high-performance pipeline to automatically segment and analyze mitochondrial morphology and ultrastructure from low to high-resolution images acquired with Transmission Electron microscopy (TEM). Our tool was designed to analyze mitochondria from multiple tissues and multiple species (Fish, Worms, Mouse, Fly & Human). The interface allows to compute a set of morphometrics and texture measurements, and provides a list of graphs for optimizing data visualization. An additional machine learning (ML) module with predictive analytic tools allows determining how any given experimental condition would impact on mito-metrics

EMito-Metrix app was written by Emmanuel Doumard, Elena Morin, Jean-Philippe Pradère and Mathieu Vigneau from the RESTORE Institute.

This section describes EMito-Metrix instructions for running and output descriptions

🔔 Important Notes 🔔

- **We strongly recommend all users to carefully follow all instructions in this Instruction manual or FAQ sections as described in the EMitometrix website (link)**
- **EMito-metrix app is an open Access tool to support all scientists in the field, and help our community to move forward. But in any case, Emito-Metrix staff assumes no responsibility, for errors, omissions, data interpretation, or any actions taken based on the utilization of our application.**
- **If you use in your publication results generated from our application, please cite our work as followed: "EMito-Metrix enables automated evaluation of mitochondria morphology across species" Morin E, Doumard E et al, Nature Metabolism, 2025. // doi10.1038/s42255-025-01400-z https://www.nature.com/articles/s42255-025-01400-z**
- **If you use our work, be care about the following license : EMito-Metrix License – research purposes restricted**

- TABLE OF CONTENTS

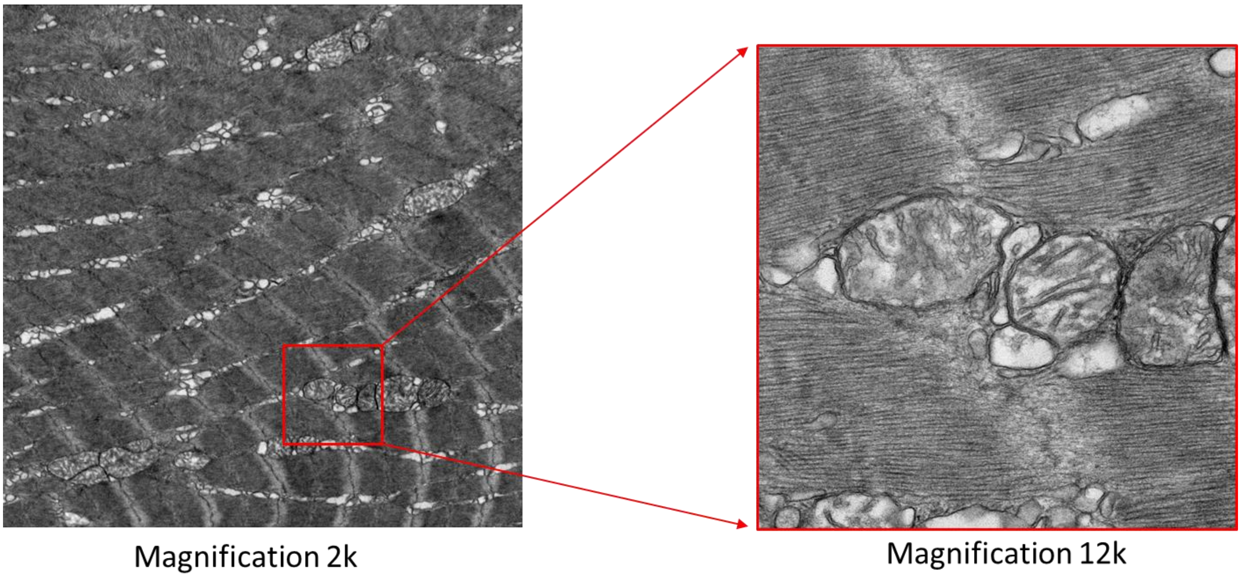
WARNINGS, ADVICES & PREREQUISITES

1- WARNINGS & ADVICES ABOUT ELECTRON MICROSCOPY IMAGES

- Does EM magnification modify mitochondrial measurements?

In electron microscopy (EM) mitochondrial analysis, panoramic images enable the visualization of the tissue but not the details of each mitochondrion. Larger images are useful for gaining finer mitochondria measurements, such as Cristae's orientation, parallelism, and Cristae's quantity within the mitochondria.

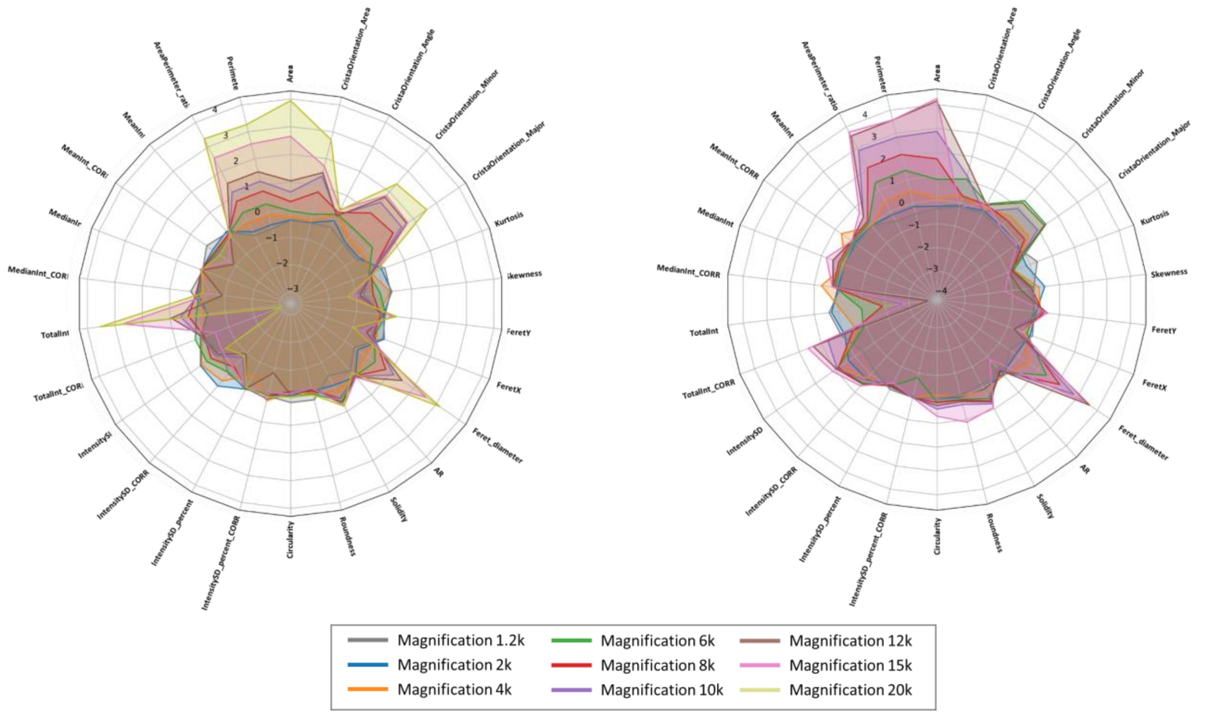
In optical microscopy, the resolution of images is closely related to the magnification used during acquisition. The higher magnification, the greater resolution for morphology and ultrastructure details (see figure below).



EM image of the same skeletal muscle biopsy of Zebrafish, using either a magnification of 2k (left panel) or 12k (right panel). The higher magnification, the greater resolution of mitochondrion structure.

Before acquiring your EM images, we recommend that you adjust your magnification to the accuracy you would like to obtain for your morphological and ultrastructure mitochondrial measurements.

Modifying magnification during the acquisition will affect the resolution of EM images, i.e. the resolution of mitochondrial features. Therefore, when running EMito-Metrix analysis, this magnification variation will result in variations of some mitochondrial measurements – especially morphological measurements -, as shown in the following figure.

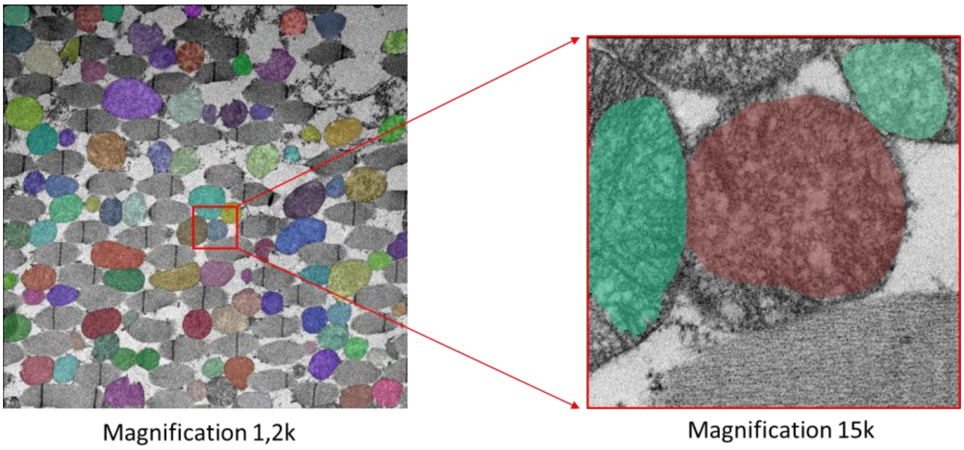


Impact of EM magnification on mitochondria metrics measurements from images of Mouse's skeletal muscles (left panel) and Drosophila's skeletal muscle (right panel)

If you plan to analyze and compare one-by-one several conditions, we highly recommend that you use the same magnification so that you can compare the corresponding morphological measurements

- **Does EM magnification affect mitochondrial segmentation accuracy?**

While using a high magnification may be necessary to obtain high-resolution images of mitochondria, it is worth noting that precision and sensitivity of the model used for mitochondria segmentation is highly dependent on mitochondria's environment, i.e. the type of tissue imaged (see below).

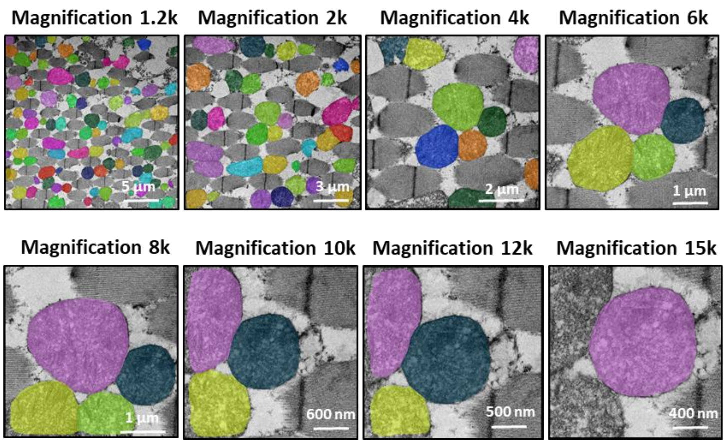


EM image of the same skeletal muscle biopsy of Drosophila, acquired at magnification of 1.2k (left panel) or 15k (right panel). For each case, we have projected mitochondria segmentation on the raw EM image. We observe an erroneous segmentation from the 15k magnification, which is due to the absence of tissue's context

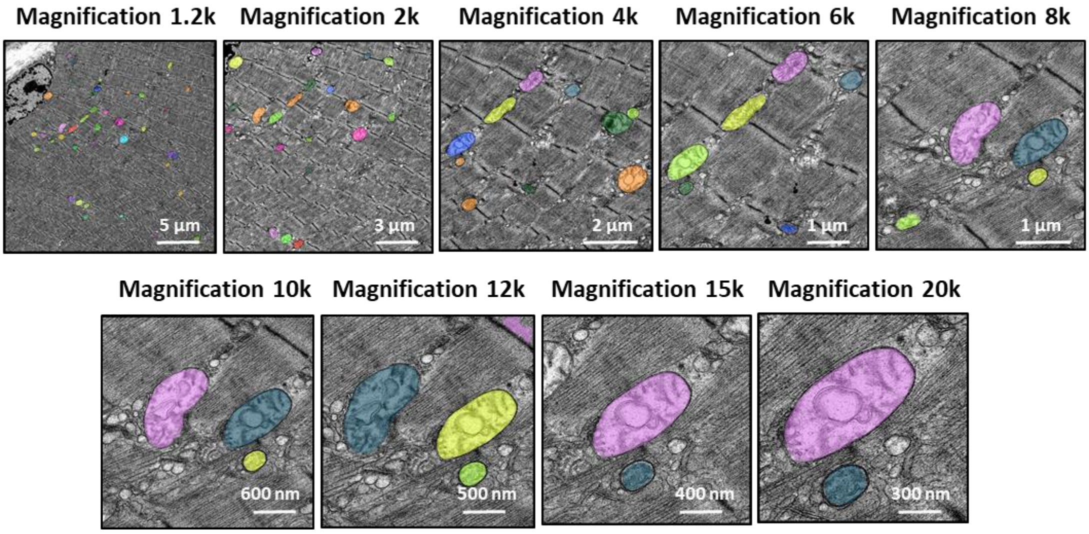
When setting both magnification and tissue's region to acquire, we recommended that you choose an area containing several mitochondria (at least 2 mitochondria) associated with their tissue environment. This should guarantee better mitochondria detection (see below).

We tested if our fine-tuned GM model was working at high magnification. Specie per Specie, we compared mitochondria segmentation from 1,2k to 20k resolution (see figures below). We found that mitochondria segmentation was working very well but with a variable accuracy depending on the specie. Thus, we established that our model works at maximum 15k images for Fly, 12k images for Z-Fish and 20k images for Mouse.

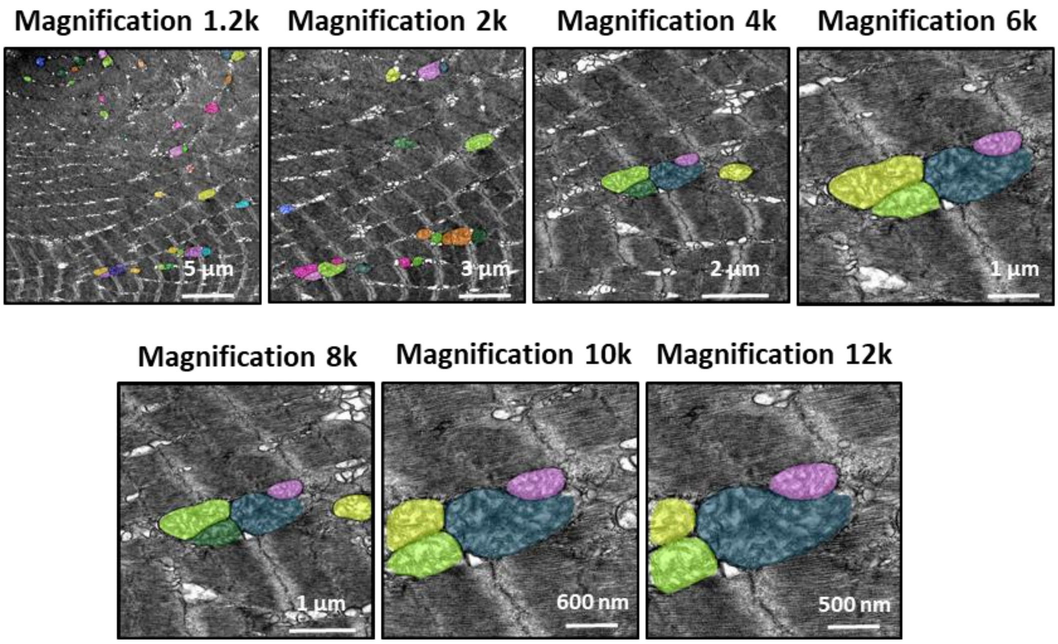
- **Impact of resolution on DROSOPHILA mitochondria segmentation**



- **Impact of resolution on MOUSE mitochondria segmentation**



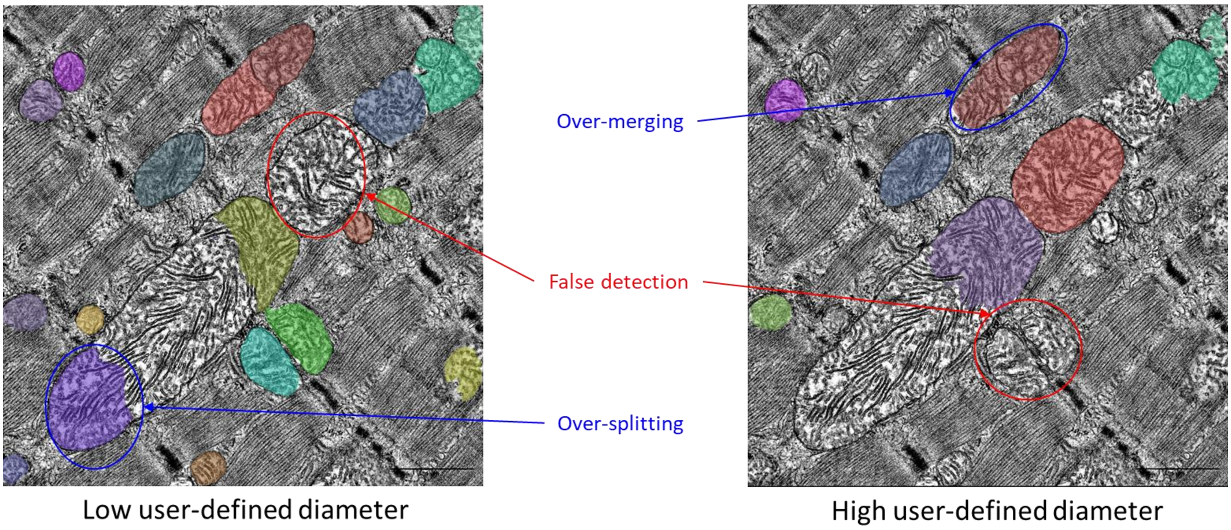
- **Impact of resolution on ZEBRAFISH mitochondria segmentation**



- **Does heterogeneity of mitochondria's size affect segmentation accuracy ?**

The Cellpose neural network used in our EMitoMetrix application has been separately trained on images from tissues and different species, resulting in so many different species-specific models. For each model, we annotated mitochondria with various diameter; however, using Cellpose neural network in an automatic way needs a user-defined mitochondria diameter (in pixel) as an input for segmentation.

User defines the size value as the average diameter of mitochondria from a single image, or from an image-by-image basis. Changing the diameter will change the results of the algorithm outputs (i.e. the objects segmented). When the diameter is set smaller than the true size, then the neural network model may over-split mitochondria. Similarly, if the diameter is set too big, then the neural network model may over-merge mitochondria. In the Emitometrix interface, user has to define a main diameter per image (or per condition). Therefore, if the tissue's region imaged contains mitochondria with strong size heterogeneity (i.e. large and small mitochondria in the same image), the neural network model may detect mitochondria with a poor accuracy (for example an over-splitting of large mitochondria and/or an over-merging of small mitochondria, as shown below).



Projection of EM image of skeletal muscle biopsy of Mouse with mitochondria segmentation map (color-coded). We used either a low value (left panel) or a high value as a user-defined diameter for mitochondria detection*

*Images from CBM (CSIC-UAM), UAM University - Laura Formentini.

To overcome this issue, the user has the option to define a range of mitochondrial diameters (a minimum and a maximum value) in addition to the main value. This feature is optional, meaning that the user can choose to set only one value (either minimum or maximum), both, or neither.

- Does image quality/contrast affect mitochondrial segmentation accuracy?

In conventional optical microscopy, the quality of the biological information contained within an image depends on multiple factors acting throughout the data lifecycle. It relies not only on the algorithms and procedures employed during the image processing and analysis stages but, more critically, on upstream steps such as sample preparation and image acquisition. Although numerous analytical tools currently enable restoration or correction of image intensities, a low-contrast or low-resolution image will yield poor-quality information, regardless of its nature (quantitative, morphological, or qualitative). This issue is particularly pronounced when attempting to identify and analyze the morphology of fine and highly resolute structures such as mitochondria. For these reasons, it appears essential for experimenters to rigorously control and validate a series of prerequisites during sample preparation and image acquisition prior to EMitoMetrix analysis, as discussed here (Culley S, et al., 2023).

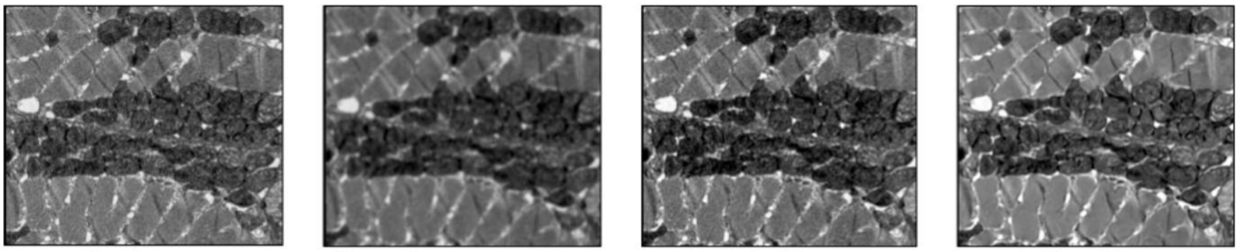
For the training of all Specialist and Generalist models, we selected images acquired with different magnifications (from 1.2k to 20k). We ensured that each selected image had high-contrast and high-resolution, thus excluding images with a low signal-to-noise ratio.

For all these reasons, the segmentation proposed in our EMitoMetrix tool may present certain reservations - regarding its precision and sensitivity - in two specific cases:

- *using low-contrast or low-resolution images : difficulty to differentiate mitochondrial cristae from the background, or badly separating mitochondria close to each other.*
- *using images acquired with very high magnification (>15/20k): over-segmentation of mitochondria due to higher-resolute morphology/cristae than the one used for training, or insufficient tissue context (see here).*

To partially correct for these limitations highly dependent on image quality, EMitoMetrix has been enhanced with three restoration/filter functionalities that enable users to correct images prior to segmentation:

- *debluring which enable to restore low resolute images*
- *denoising which enable to restore images with low signal-to-noise ratio.*
- *Gaussian filter which enable to smooth image, enhancing segmentation for higher resolute images*



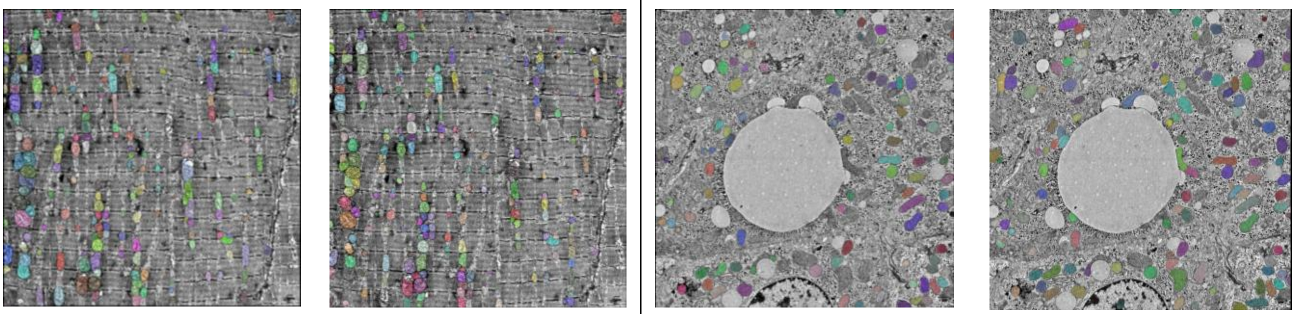
Raw image | Filtered image *(Gaussian blur)* | Denoised image | Deblured image

These functionalities are already implemented in the Cellpose 3.0 tool, as described here (Stringer C., et al. 2024).

- **Should I use Generalist or Specialist Model for Mitochondria Segmentation?**

Each specie-specific model has been trained using images of the different tissues from single specie (i.e. ZebraFish, Mouse, Fly, Human, Killifish or C-Elegans). This implied that dataset used for each model training had a high homogeneity of tissue's environment. On this opposite, we used images of skeletal muscle from several species (i.e. ZebraFish, Mouse, Killifish, C-Elegans and Human) to train our Generalist model, that is dataset with a very high heterogeneity of tissue's environment. Because of this heterogeneity, and depending on the tissue and/or specie you use, you may have a better mitochondria detection using the Generalist model than the Species-specific models (see below).

Mouse Skeletal muscle | **Mouse Liver**



***Generalist model*: 201 objects** | ***Mice-Specific model*: 230 objects** | ***Mice-Specific model* : 96 objects** | ***Generalist model* : 117 objects**

Projection of EM image of skeletal muscle biopsy (left panels) and liver biopsy (right panels) of Mice with mitochondria segmentation map (color-coded). For each tissue, we used either Generalist or Mice-specific model as an input for Cellpose mitochondria detection. The number of detection is indicated for each condition*

* Images from CBM (CSIC-UAM), UAM University - Laura Formentini

For better precision and sensitivity of detection, we recommend testing both Generalist and Specific models on your dataset, before starting EMitoMetrix application

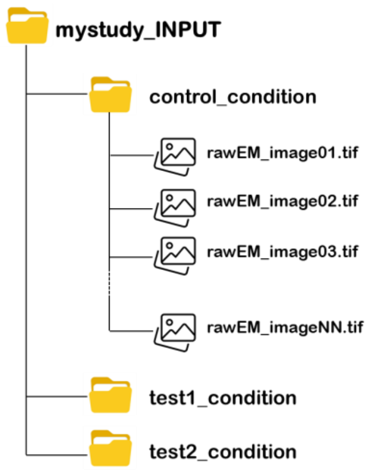
We do not recommend using Cellpose trained models (cyto2, nucleus, cyto), as these models have not been trained on EM images.

2- APPLICATION EXPECTATIONS

- Expected input files & folders

EMito-Metrix application allows: 1- analyzing mitochondrial morphology and ultrastructure in EM images **from multiple experimental conditions**; 2- **comparing these conditions one-by-one** in a single (and same) analysis.

Before running such analysis, we recommend that you organize your experimental conditions and input files (i.e. raw EM images) as shown below:



- Save all input files and conditions to be analyzed and compared in a unique parent folder (*mystudy_INPUT)*
  No preferences regarding the location of mystudy_INPUT folder

- *mystudy_INPUT* must contain as many folders as conditions to compare/analyze
  Example: "control_condition vs test_condition

- *mystudy_INPUT* must contain nothing but the condition folders to compare/analyze.
  No other files or folders saved in mystudy_INPUT folder

- Each sub-folder must contain all input (raw) EM images generated from your condition.
  Condition folders <u>must contain nothing but the input (raw) images</u>

- Folder name and input files name must respect the following rules:
  no special characters: []!#$%&'()/,:;<=>?@^`{|}~]*
  no spaces, no accents
  no dot (except the one for file extension)

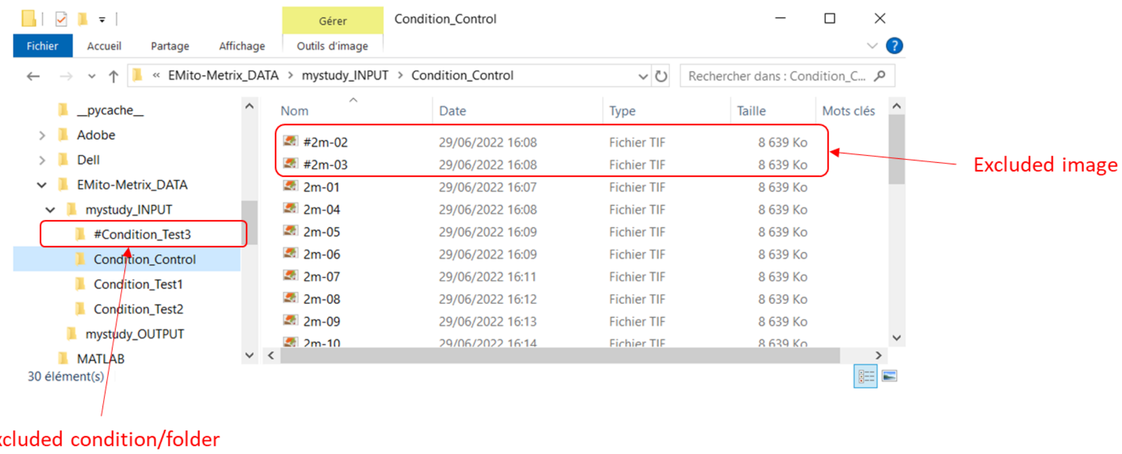
!! the name of each condition will be used and displayed in the different output generated from computation modules !!

- We recommend using the following file formats for the analysis:

tif images, tiff images, bmp images

!! we do not recommend using jpg, jpeg and png file formats, because of their poor resolution !!

To exclude conditions and/or images from the analysis, insert the character # at the beginning of the folder's/image's name you want to exclude (see example below).



- Expected output folders

Create an output folder (i.e. *mystudy_OUTPUT)* dedicated to output folders and files generated by the EMito-Metrix application. **You must organize *mystudy_OUTPUT* as below:**

- *mystudy_OUTPUT* must be empty when performing the analysis for the first time

- *mystudy_OUTPUT* name must respect the following rules:

no special characters: []!#$%&'()/,:;<=>?@^`{|}~]*

no spaces, no accents

- While running, the application will create output folders and files in *mystudy_OUTPUT* folder.

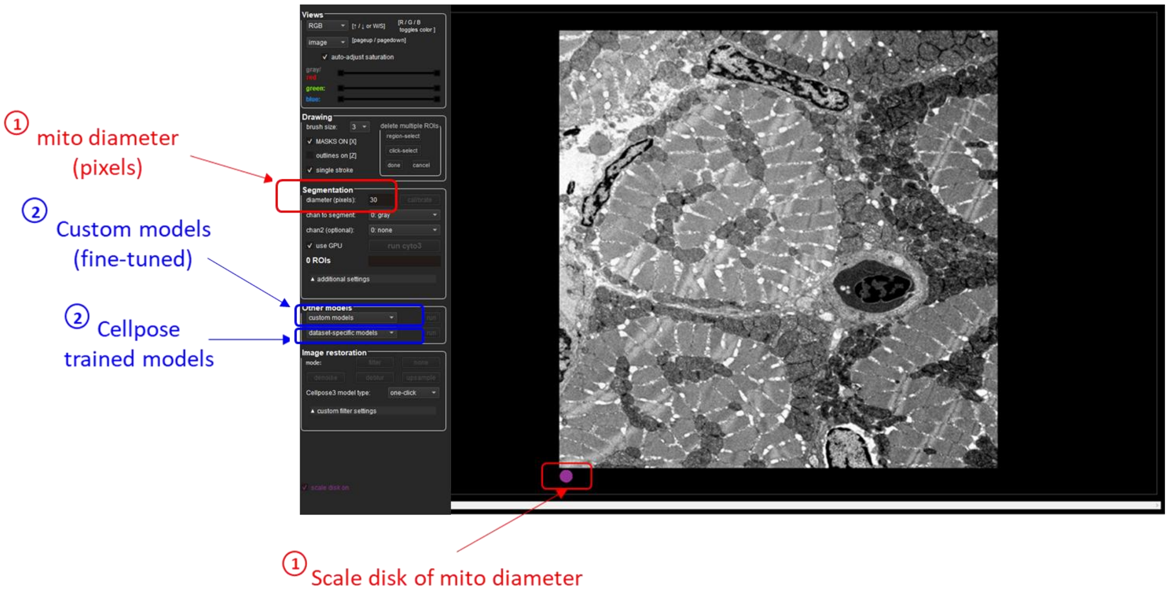
mystudy_OUTPUT must contain nothing but outputs.

During the analysis, do not modify any of the output folders and files created in the OUTPUT folder, which may abort the application running

- Mitochondria segmentation settings

Before running EMito-Metrix application, you have to estimate **mitochondria** size as well as **trained model** for mitochondria segmentation. To set these parameters, **use Cellpose graphical user interface** by clicking on the Cellpose shortcut (see here for Mac user instructions).

In Cellpose window, drag and drop your input image to analyze. Here is an example of the Cellpose graphical interface and its functionalities:



Using Cellpose application, estimate the following segmentation settings:

- **Mitochondria size (in pixels):** defined as the average diameter of mitochondria to detect in the image (see here for more details about diameter definition)

  Use the purple circle (at the bottom of the view) as a scale disk of the user-defined diameter value

- **Trained model to use for automatic segmentation** (see here for more details)

  You can choose either a Cellpose trained model (cyto2, nucleus, cyto) or one of our custom Generalist or Species-specific EM models.

Once you have set both diameter and trained model, click on the *run button* to check the mitochondria segmentation. **Adjust diameter accordingly if the segmentation is not working**.

Depending on mitochondria size heterogeneity of your dataset, you may need to set your diameter value by image, by condition (single value for all images of the condition) or by study (single value for all conditions)

For more details about Cellpose GUI instructions, check out this documentation

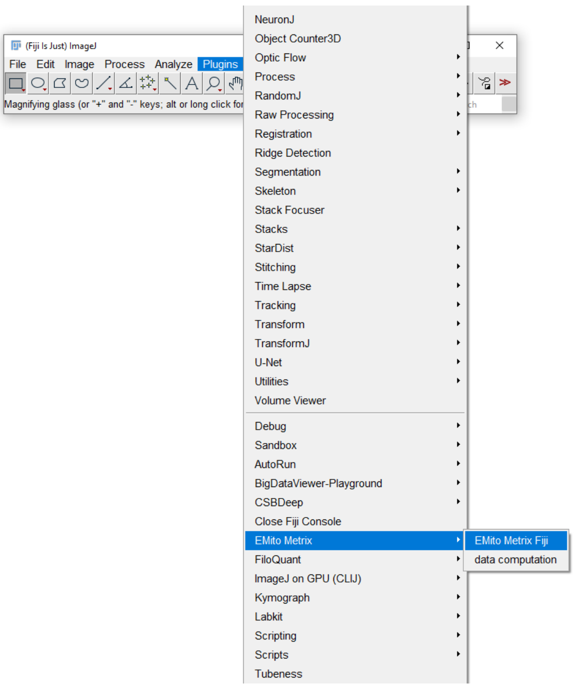
EMITO-METRIX RUNNING

HOW TO LAUNCH EMITO-METRIX APPLICATION?

EMito-Metrix setting and running are performed using Fiji application (see *here* for installation instructions).

- In your Fiji application directory, double-click the *ImageJ-win64.exe* file to start Fiji software

- Once the Fiji application has started, go to *EMito_Metrix* menu from Fiji *Plugins* menu, and start the application selecting *EMito_Metrix_Fiji,* as shown below :



- *EMito_Metrix* application was designed to meet 4 needs :
  - detecting and segmenting mitochondria in an automatic way
  - characterizing mitochondria using morphology & ultrastructure measurements
  - generating graphs and data distributions for mitometrics display
  - generating predictive models based on mitochondrial measurements

EMito-Metrix application allows launching each feature sequentially

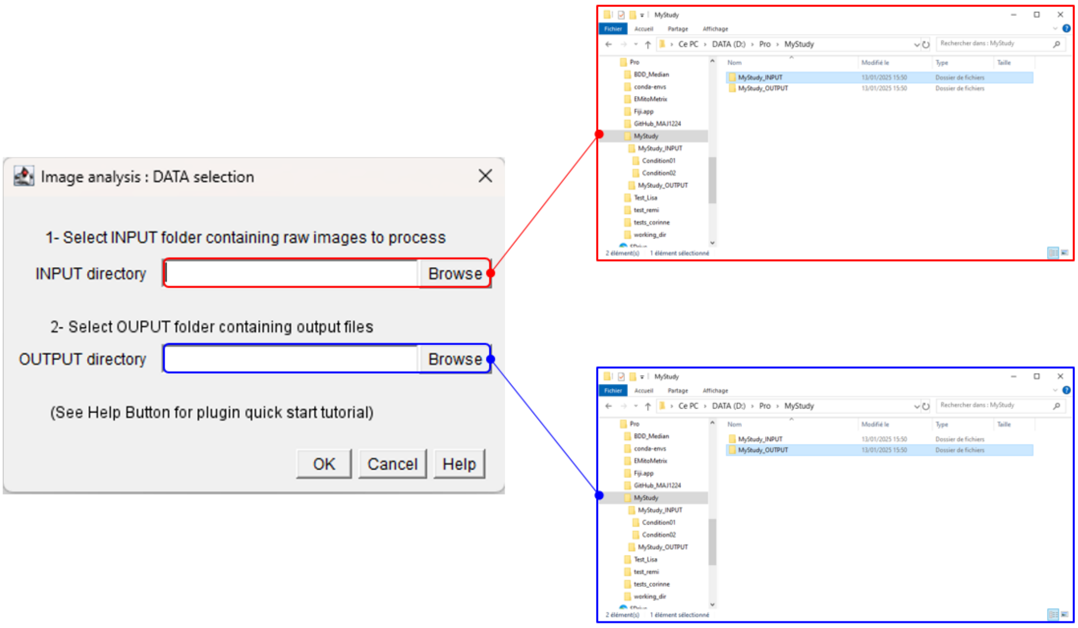
1- FIRST STEP: AUTOMATIC MITOCHONDRIAL SEGMENTATION

- A/ Setting INPUT and OUTPUT folders

If the EMito-Metrix application installation and setting is correct, the next window will invited you to set these following parameters:

- **INPUT Directory**: specify root directory containing condition folders and input files
  Example: C:\Users\username\Documents\EMito-Metrix_DATA\mystudy_INPUT

- **OUTPUT Directory**: specify root directory used to save output files, images and measurements
  Example: C:\Users\username\Documents\EMito-Metrix_DATA\mystudy_OUTPUT

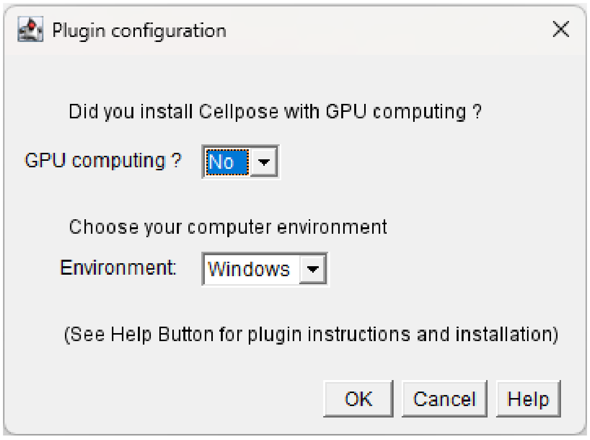


The application will check the validity of INPUT and OUTPUT folders and files *(see here for detailed instruction)*

If INPUT and/or OUTPUT folders and files are not valid, the application execution will abort and invite you to check INPUT and OUTPUT folders/files.

- B/ EMito-Metrix plugin configuration

If the INPUT and OUTPUT folders are correct, you will have to set these following parameters:



- **GPU computing**: specify if you have installed a GPU version of Cellpose
  Use Gpu if you have a compatible Nvidia GPU with CUDA installed (see here)

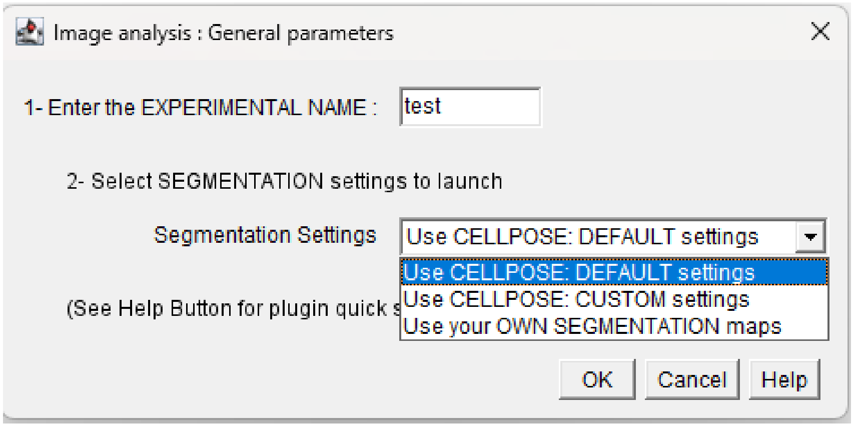
- **Environment**: specify your computer environment used for EMito-Metrix running
  Compatible with Windows, MacOS and Linux environments

The application will check the validity of your specified Cellpose and Emitometrix virtual environments *(see here for installation instruction).*

If virtual environment folders are not valid, the application will abort and invite you to check it

- C/ Selecting segmentation method

If virtual environments are correct, you will have to define the following settings:



- Experimental / protocol name

String of characters

- **Select Segmentation settings to use**

- Use CELLPOSE: DEFAULT settings

All parameters - image preprocessing and segmentation parameters - are set to default mode

--> if selected, go to "G/ Setting live image display"

- Use CELLPOSE: CUSTOM settings

Manually setting general, image preprocessing and segmentation parameters

--> if selected, go to "D/ Setting general parameters & image preprocessing"

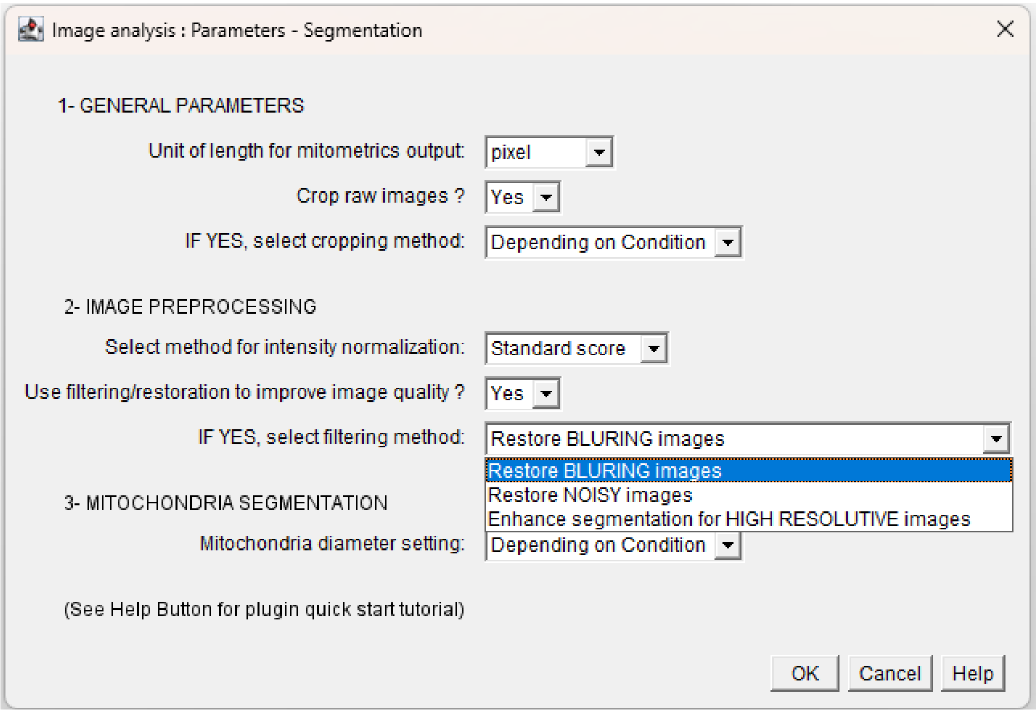
- Use your OWN SEGMENTATION maps

Importing your own segmentation maps generated outside of our pipeline

--> if selected, go to "F/ Automatic mitochondria segmentation : OWN SEGMENTATION"

- D/ Setting general parameters & image preprocessing

If you have selected "Use CELLPOSE: CUSTOM settings" from the previous step, set the general, image preprocessing and segmentation parameters, as shown below:

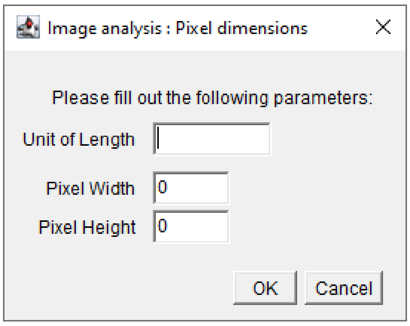


- **General parameters**

- **Unit of length**: unit used for morphometrics measurements

  pixel or calibrated

If you select *calibrated*, you will have to define *pixel width*, *pixel height* and *unit of length*.



- **Crop raw images**: select a region of interest (ROI) from your EM images

  Yes or No

- **Cropping method:** if you select *yes*, set the level of definition of the ROI

  Depending on condition: manually define the same ROI for all images of the same condition
  Depending on images: manually define a specific ROI for each image and each condition

- **Image preprocessing**

- **Normalization method**: Raw EM images are normalized on amplitude to attenuate global gray level variations observed between images, which may be related to variations in the amount of staining agent.

  min-max scaling: consists in rescaling the range of features to scale the range in [0, 1]. This method is preferred when your data does not follow a normal distribution. Does not handle outliers well.

  standard score: values (x) are centered on mean (mean(x)) with a unit standard deviation (sd(x)). This method is preferred when your data follows a normal distribution. Handles outliers well.

$$x' = \frac{x - \min(x)}{\max(x) - \min(x)}$$

Min-max scaling

$$x' = \frac{x - \mathrm{mean}(x)}{\mathrm{sd}(x)}$$

Standard score

- **Image filtering/restoration**: To partially correct for the limitations depending on image quality, EMitoMetrix has been enhanced with restoration/filter functionalities that enable users to correct images prior to segmentation

  Yes or No (default value)

- **Image filtering/restoration method**: If you select *Yes* from the previous setting, choose the filtering method

  Restore BLURING images: enable to restore low resolute images

  Restore NOISY images: enable to restore images with low signal-to-noise ratio

  Enhance segmentation for HIGH RESOLUTIVE images Gaussian filter which enable to smooth image intensities, enhancing segmentation for higher resolute images

- **Mitochondria segmentation**

Mitochondria diameter: select the level of mitochondria size variability (decide if mitochondria size varies across conditions or/and images or is the same for all conditions and images)

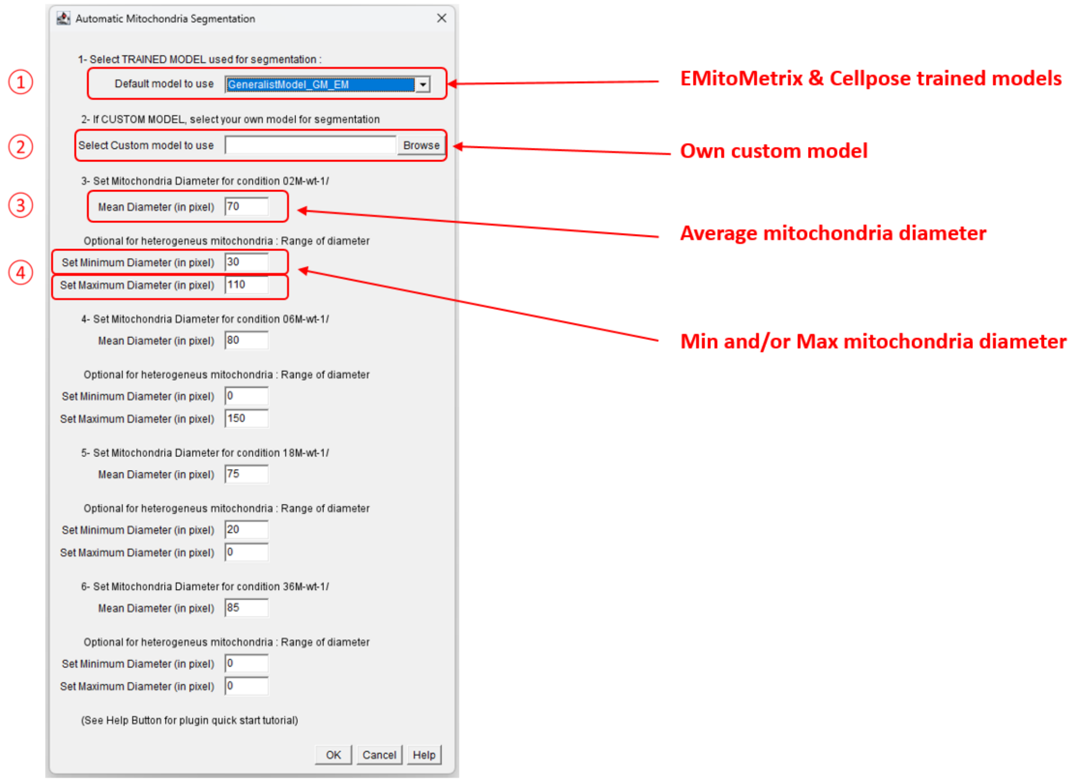
"Depending on condition": manually setting the same diameter for all images of the same condition
"Depending on images": manually setting mitochondria diameter image by image

- E/ Automatic mitochondria segmentation : CUSTOM SETTINGS

You have selected "CUSTOM SEGMENTATION" from the previous step

Depending on the options set in the previous section, you will have to define the following mitochondria segmentation settings:

- **Segmentation settings depending on CONDITION**



- Default model used for mitochondria segmentation:

Cellpose models (cyto, cyto2, cyto3 or nucleus), EMito-Metrix trained models (Specialist or Generalist models) or custom models (your own trained/fine-tuned model)

- If custom model: specify your own custom trained/fine-tuned model

Select your trained/fine-tuned model file with a dialog box

- Mean Mitochondria diameter (in pixels): average size of mitochondria to detect in the image (see here for detailed instruction)

- Minimun and Maximum Mitochondria diameter (in pixels): option to define a range of mitochondrial diameters (a minimum and a maximum value) in addition to the main value. This feature is optional, meaning that the user can choose to set only one value (either minimum or maximum), both, or neither (see here for detailed instruction).

- F/ Automatic mitochondria segmentation : OWN SEGMENTATION

You have selected "OWN SEGMENTATION" from the previous step

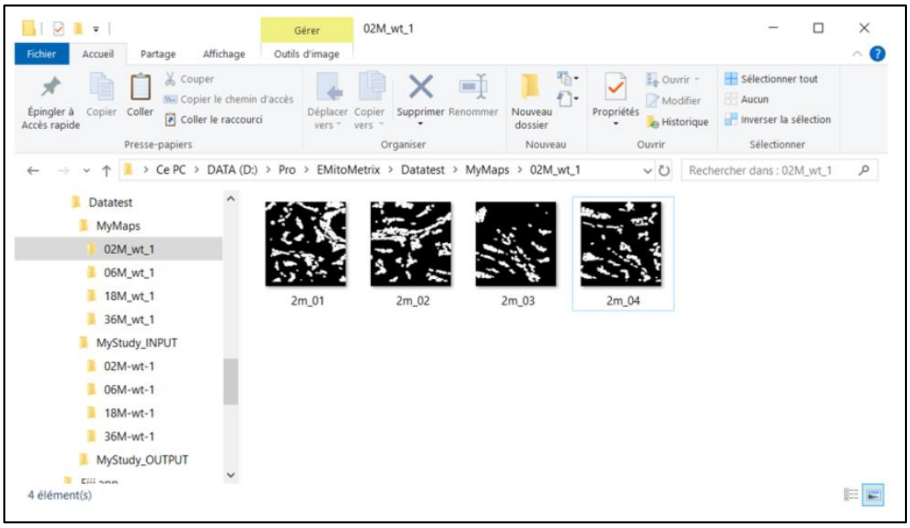
We have added, in EMitoMetrix tool, the option for the user to import their own segmentation maps (segmentation performed outside of our pipeline, Empanada for example or with own-trained model), and putting back into the EMitoMetrix pipeline in order to perform data analysis, mitometrics visualization and data prediction. Here are the instructions to follow carefully to import and analyze your own segmentation maps in the EMitoMetrix tool.

Before starting the analysis, your data must be organized as follows:

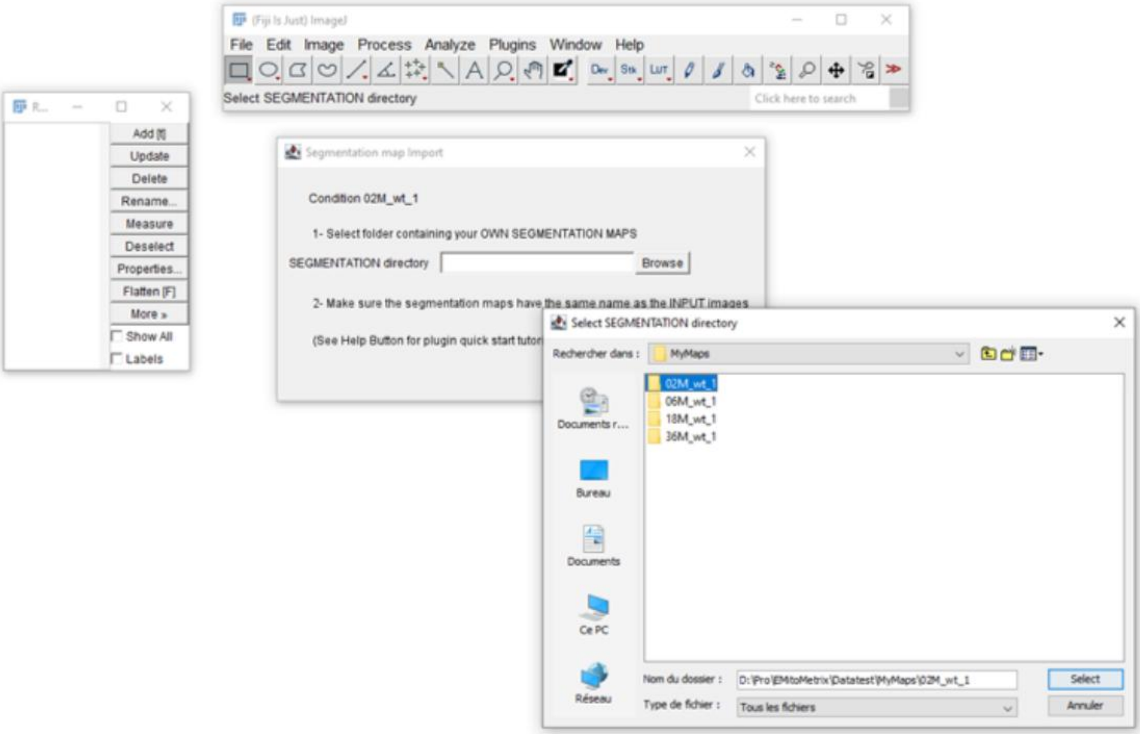
- An INPUT folder containing your raw TEM images (see here for more details).

- An empty OUTPUT folder that will contain the output data.

- For each condition, a MAPS folder containing the segmentation maps. Each MAPS folder should contain the segmentation maps only. The filename of each map should match that of the corresponding raw image.



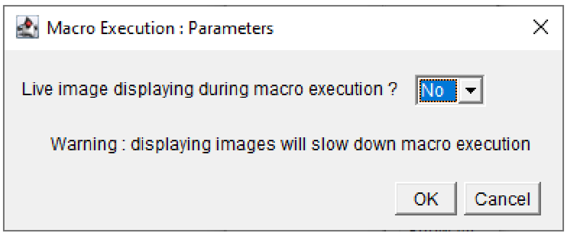
Once you have selected *"Automatic mitochondria segmentation: OWN SEGMENTATION"* from the previous step, the *Segmentation map import* window appears. For each condition, indicate the MAPS folder containing the segmentation maps (as shown below). These segmentation maps will be copied into the corresponding *Cellpose_mask* folder in the *OUTPUT/image_output* directory



- **G/ Setting live image display**

You have selected "CUSTOM SEGMENTATION" or "DEFAULT SEGMENTATION" from the previous step

A batch mode is proposed, that allows masking image display during application execution.

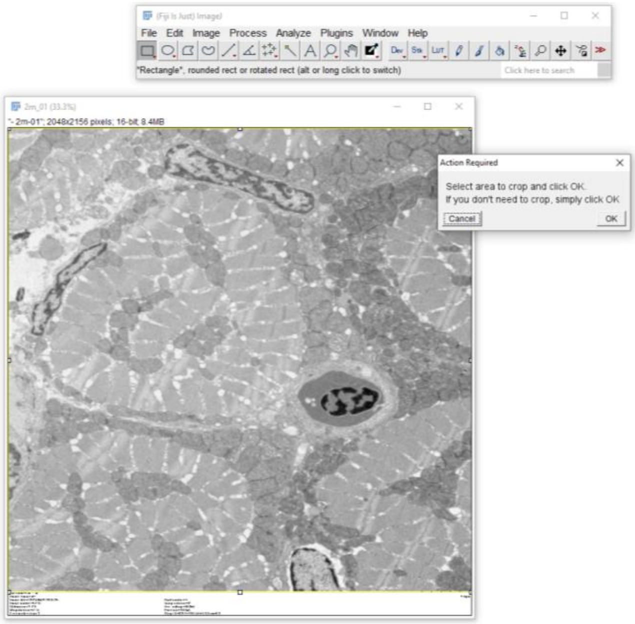


Selecting *live image displaying* (*Yes* settings) may slow down the application execution

- **H/ Setting raw EM images cropping**

You have selected "CUSTOM SEGMENTATION" from the previous step

Before running mitochondria segmentation and morphological analysis, you can crop your input EM images for each condition separately:



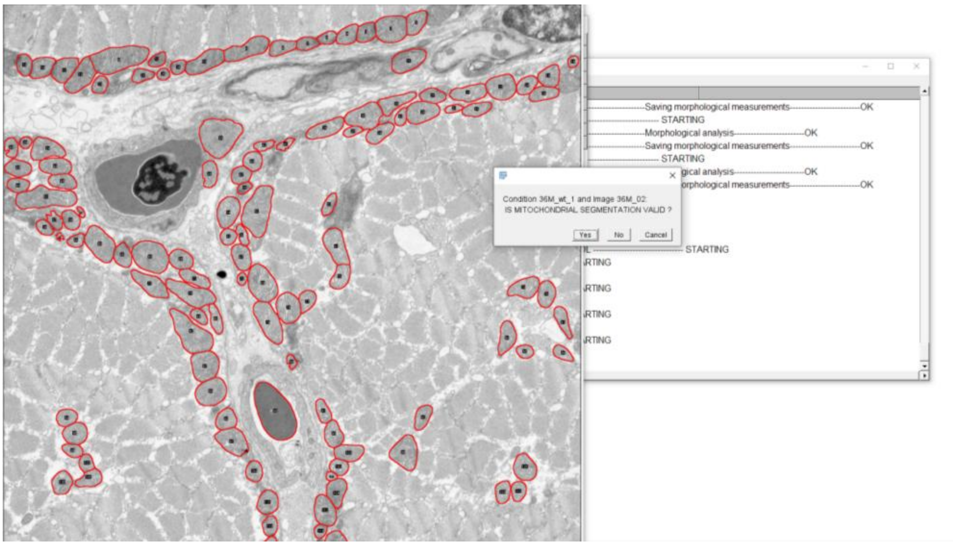
- **<u>If it is necessary</u>, click on *Yes***. Then, **select an area of interest to keep** in your image using the *rectangle tool* from Fiji application. Once the region of interest is selected, click *ok*. The **application will crop all images of the condition, <u>in the same way</u>**.

- **<u>If it is not necessary</u>, click on *No***. The application will continue without cropping images.

Then mitochondria segmentation is starting image per image and condition per condition; you will be able to follow the processing on the "LOG" Window.

- I/ Validating Mitochondria segmentation

Once mitochondria detection is done, validate or invalidate mitochondria segmentation for each input EM image of each condition, as shown below:



- **<u>Click on *No* button</u>** if you consider that mitochondria **detection is not good**.

- **<u>Click on *Yes* button</u>** if you consider that mitochondria **detection is good**.

Once segmentation validation is done, the application will continue using all images and all mitochondria but the invalid ones

Once segmentation is done, the user will be able to restart EMitoMetrix to proceed to the next steps (analysis, data display, and data prediction).

2- SECOND STEP: MITOMETRICS MEASUREMENTS

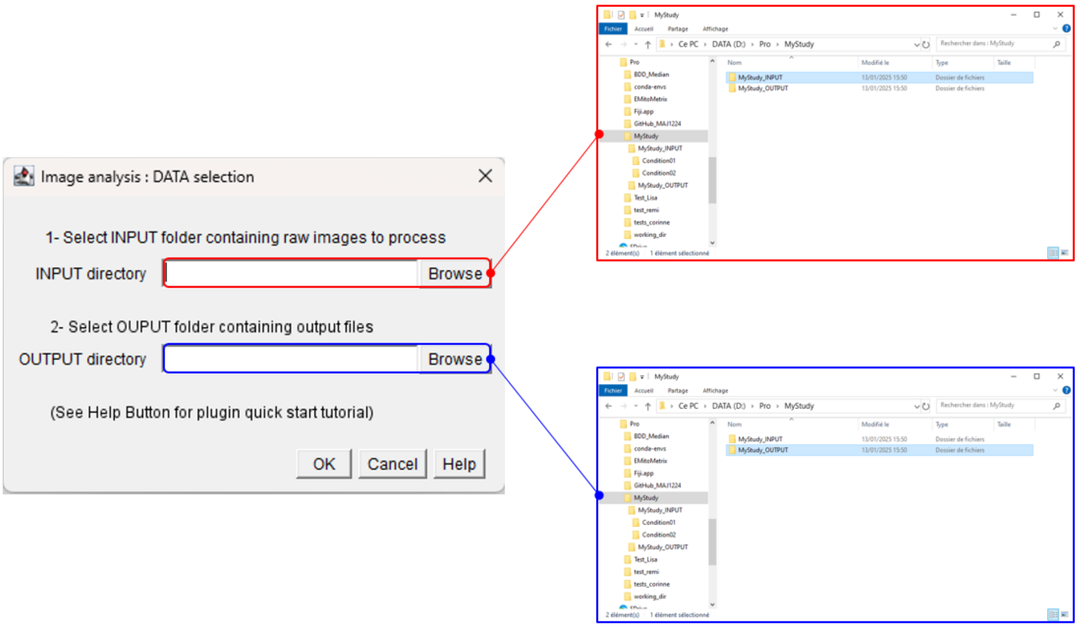
In Fiji application, go to *EMito_Metrix* menu from *Plugins* menu, and start the analysis selecting *EMito_Metrix_Fiji.*

- Setting INPUT and OUTPUT folders

If the EMito-Metrix application installation and setting is correct, the next window will invited you to set these following parameters:

- **INPUT Directory**: specify root directory containing condition folders and input files
  Example: C:\Users\username\Documents\EMito-Metrix_DATA\mystudy_INPUT

- **OUTPUT Directory**: specify root directory used to save output files, images and measurements
  Example: C:\Users\username\Documents\EMito-Metrix_DATA\mystudy_OUTPUT



The application will check the validity of INPUT and OUTPUT folders and files *(see here for detailed instruction)*

If INPUT and/or OUTPUT folders and files are not valid, the application execution will abort and invite you to check INPUT and OUTPUT folders/files.

If INPUT and OUTPUT folders are correct, the EMito-Metrix application will compute mitometrics measurements (morphology and texture) from the segmentation maps computed in the previous step (see here for a detail description of mitometrics). Mitometrics computation will start image per image and condition per condition; you will be able to follow the processing on the "LOG" Window.

File Edit Font Results

	Area	Mean	StdDev	Min	Max	X	Y	XM	YM	Perim.	BX	BY	Wi
5	3432	0.005	0.168	-1.205	0.578	50.962	20.519	80.342	19.878	242.099	0	0	98
6	739	255.000	0.000	255.000	255.000	64.500	64.500	64.500	64.500	584.247	43	19	43
7	3838	-0.172	0.469	-1.454	1.768	1325.227	37.529	1316.470	46.561	235.202	1291	3	74
8	3838	3.199E-4	0.160	-0.563	0.577	34.227	34.529	214.612	213.771	235.202	0	0	74
9	1079	255.000	0.000	255.000	255.000	64.500	64.500	64.500	64.500	950.202	15	4	99
10	1436	-0.340	0.337	-1.636	0.653	1219.674	44.604	1223.265	42.654	140.681	1197	24	45
11	1436	-3.396E-4	0.158	-0.736	0.456	22.674	20.604	200.648	-22.372	140.681	0	0	45
12	305	255.000	0.000	255.000	255.000	32.500	32.500	32.500	32.500	315.044	6	1	53
13	640	-0.804	0.251	-2.716	-0.121	157.944	43.288	158.170	42.609	93.499	144	28	28
14	640	4.707E-5	0.033	-0.286	0.104	13.944	15.287	-0.681	17.187	93.499	0	0	28
15	786	255.000	0.000	255.000	255.000	16.113	15.991	16.113	15.991	566.791	0	0	32
16	8176	0.072	0.481	-2.423	1.835	576.302	105.762	532.456	110.919	387.524	497	62	15
17	8176	9.826E-4	0.162	-0.656	0.613	79.302	43.762	25.332	46.538	387.524	0	0	15
18	3808	255.000	0.000	255.000	255.000	128.500	128.399	128.500	128.399	2718.208	33	0	19
19	2566	-0.455	0.315	-1.690	0.791	666.034	89.084	667.122	91.686	245.019	631	62	69
20	2566	2.713E-4	0.154	-0.500	0.504	35.034	27.084	7.194	-71.462	245.019	0	0	69
21	767	255.000	0.000	255.000	255.000	64.500	64.500	64.500	64.500	694.271	17	38	95
22	3291	-0.293	0.383	-2.221	1.494	1458.518	95.332	1459.128	90.204	214.619	1424	66	73
23	3291	0.001	0.165	-1.617	0.510	34.518	29.332	55.934	122.714	214.619	0	0	73
24	871	255.000	0.000	255.000	255.000	64.500	64.500	64.500	64.500	735.041	22	20	85
25	5794	0.313	0.503	-1.340	2.123	232.329	106.834	232.245	107.780	308.154	179	75	11

Once mitometrics analysis is done, the user will be able to restart EMitoMetrix to proceed to the next steps (data display, and data prediction).

3- THIRD STEP: MITOMETRICS DISPLAY

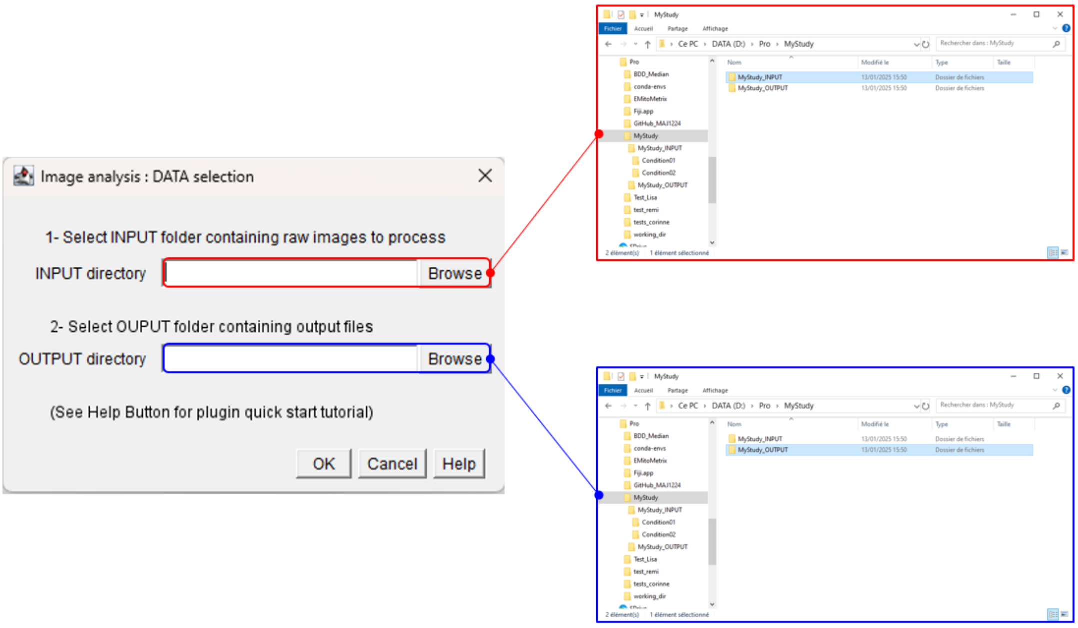
In Fiji application, go to *EMito_Metrix* menu from *Plugins* menu, and start the analysis selecting *EMito_Metrix_Fiji.*

- A/ Setting INPUT and OUTPUT folders

If the EMito-Metrix application installation and setting is correct, the next window will invited you to set these following python parameters:

- **INPUT Directory**: specify root directory containing condition folders and input files
  Example: C:\Users\username\Documents\EMito-Metrix_DATA\mystudy_INPUT

- **OUTPUT Directory**: specify root directory used to save output files, images and measurements
  Example: C:\Users\username\Documents\EMito-Metrix_DATA\mystudy_OUTPUT



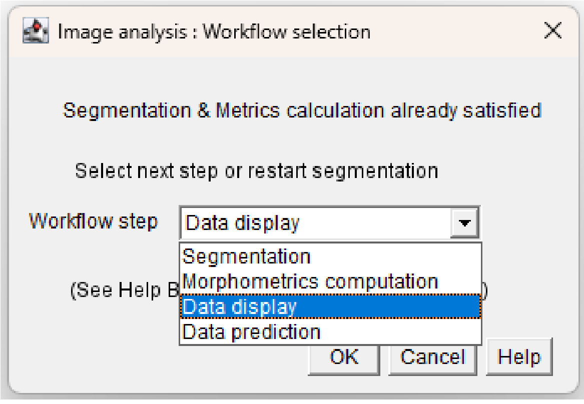
The application will check the validity of INPUT and OUTPUT folders and files *(see here for detailed instruction)*

If INPUT and/or OUTPUT folders and files are not valid, the application execution will abort and invite you to check INPUT and OUTPUT folders/files.

If INPUT and OUTPUT folders are correct, you will be able to compute graph and data distribution

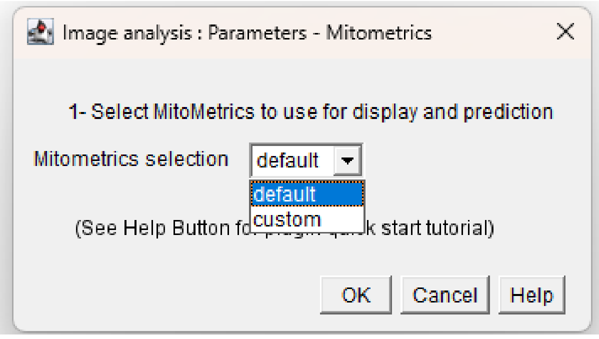
- B/ Selecting Mito-Metrix step

If INPUT and OUTPUT folders are correct, the next window will invited you to set the next step (as shown below). Select "Data display" option to generate graphs and data distribution from mitochondrial measurements



- C/ Selecting mitometrics measurements to display

Next, you will have to select morphological and texture measurements to use for data display:



- **<u>Default option</u>**: use all mitometrics for data display

- **<u>Custom option:</u>** manual selection of mitometrics to use for data display. Switch on or switch off each parameter (select at least 3 mitometrics)

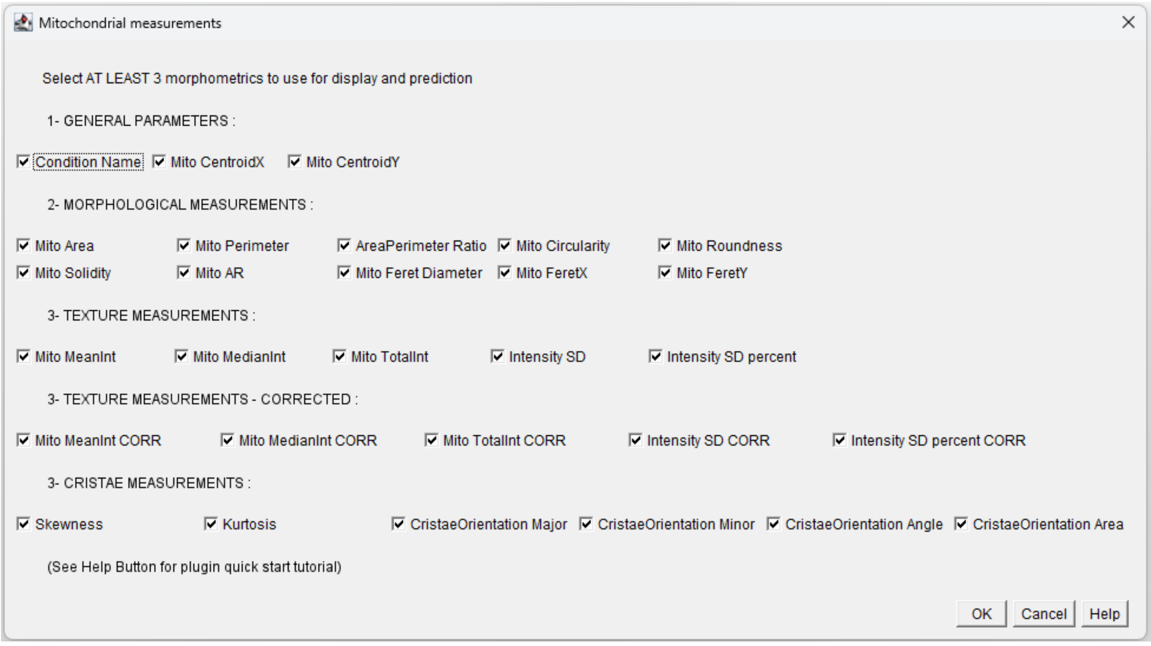
  <u>General parameters</u>: condition & images name, mitochondria labels and spatial position

  <u>Morphological measurements</u>: mitochondria shape measurements

  <u>Texture measurements</u>: ultrastructure measurements, based on distribution of normalized gray values in mitochondria

  <u>Texture measurements - corrected</u>: ultrastructure measurements, based on distribution of corrected gray values calculated in mitochondria. Corrected gray values are calculated from normalized gray values, by filtering noise with a high frequency filter (FFT)

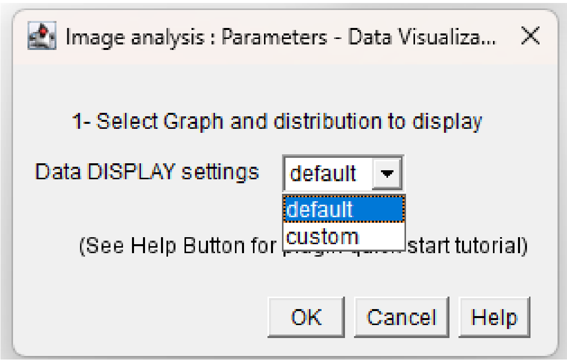
  <u>Cristae measurements</u>: measurements of cristae's organization and orientation



(see [here](#) for a detail description of mitometrics)

- D/ Selecting graphs & data distributions

Next, select graphs and data distributions used for morphological and texture visualization:



- **<u>Default option</u>**: use all graphs and data distributions
- **<u>Custom option:</u>** manual selection of graphs and data distributions (select at least 1 graph or data distribution).

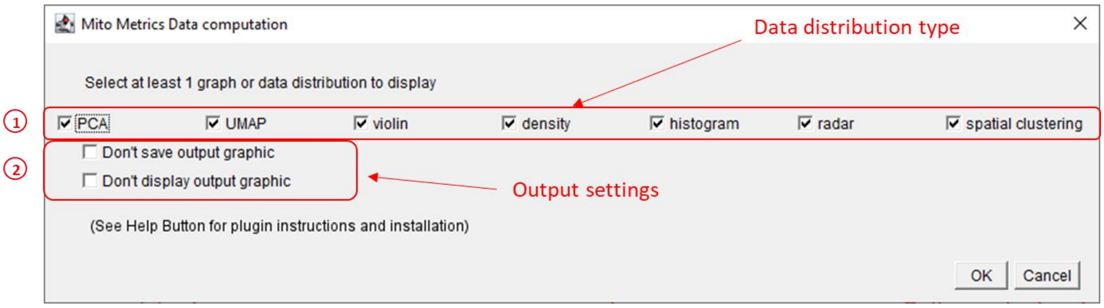
  <u>Data distribution type</u>: switch on or switch off each type (see <u>here</u> for graph description)

  <u>Others output settings: don't save output graphic</u>

  Graphs and distributions are displayed during the analysis, but <u>not saved</u> in the output folder

  <u>Others output settings: don't display output graphic</u>

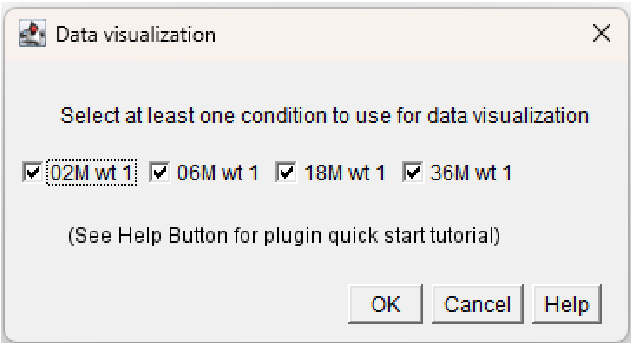
  Graphs and distributions are saved in the output folder, but <u>not displayed</u> during the analysis



(see here for a detail description of distributions)

- **E/ Selecting conditions for graphs & data distributions**

Select conditions to use for data visualization (switch on or switch off each condition). You must select at least one condition (see below). Then graph & distribution computation is starting; you will be able to follow the processing on the "LOG" Window.



Once data display is done, the user will be able to restart EMitoMetrix to proceed the next steps (data prediction/computation).

4- FOURTH STEP: DATA COMPUTATION & PREDICTION

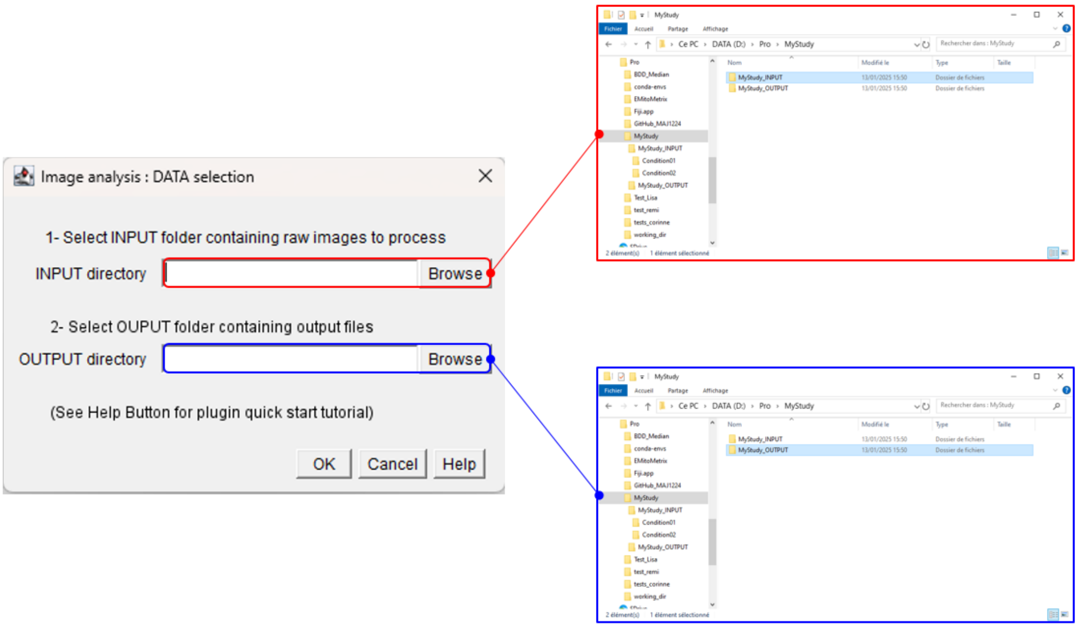
In Fiji application, go to *EMito_Metrix* menu from *Plugins* menu, and start the analysis selecting *EMito_Metrix_Fiji.*

- A/ Setting INPUT and OUTPUT folders

If the EMito-Metrix application installation and setting is correct, the next window will invited you to set these following parameters:

- **INPUT Directory**: specify root directory containing condition folders and input files
  Example: C:\Users\username\Documents\EMito-Metrix_DATA\mystudy_INPUT

- **OUTPUT Directory**: specify root directory used to save output files, images and measurements
  Example: C:\Users\username\Documents\EMito-Metrix_DATA\mystudy_OUTPUT



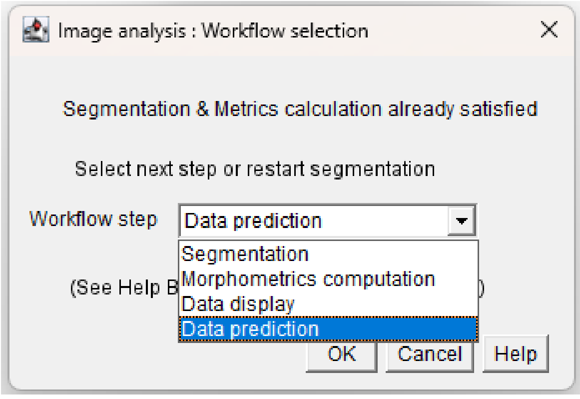
The application will check the validity of INPUT and OUTPUT folders and files *(see here for detailed instruction)*

If INPUT and/or OUTPUT folders and files are not valid, the application execution will abort and invite you to check INPUT and OUTPUT folders/files.

If INPUT and OUTPUT folders are correct, you will be able to launch data computation/prediction

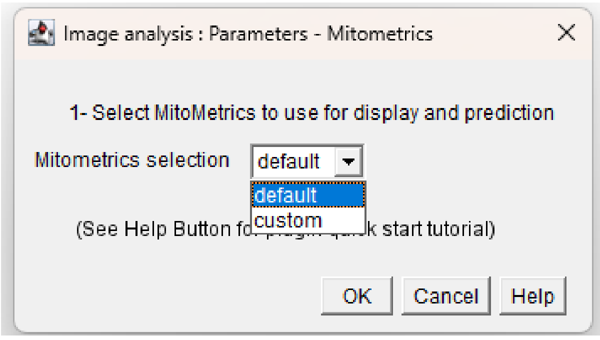
B/ Selecting Mito-Metrix step

The next window will invited you to set the next step (as shown below). Select "Data prediction" option to generate data prediction from mitochondrial measurements



C/ Selecting mitometrics measurements to display

Next, you will have to select morphological and texture measurements to use for data prediction:



- **<u>Default option</u>**: use all mitometrics for data prediction/computation
- **<u>Custom option:</u>** manual selection of mitometrics to use for data prediction/computation. Switch on or switch off each parameter (select at least 3 mitometrics)

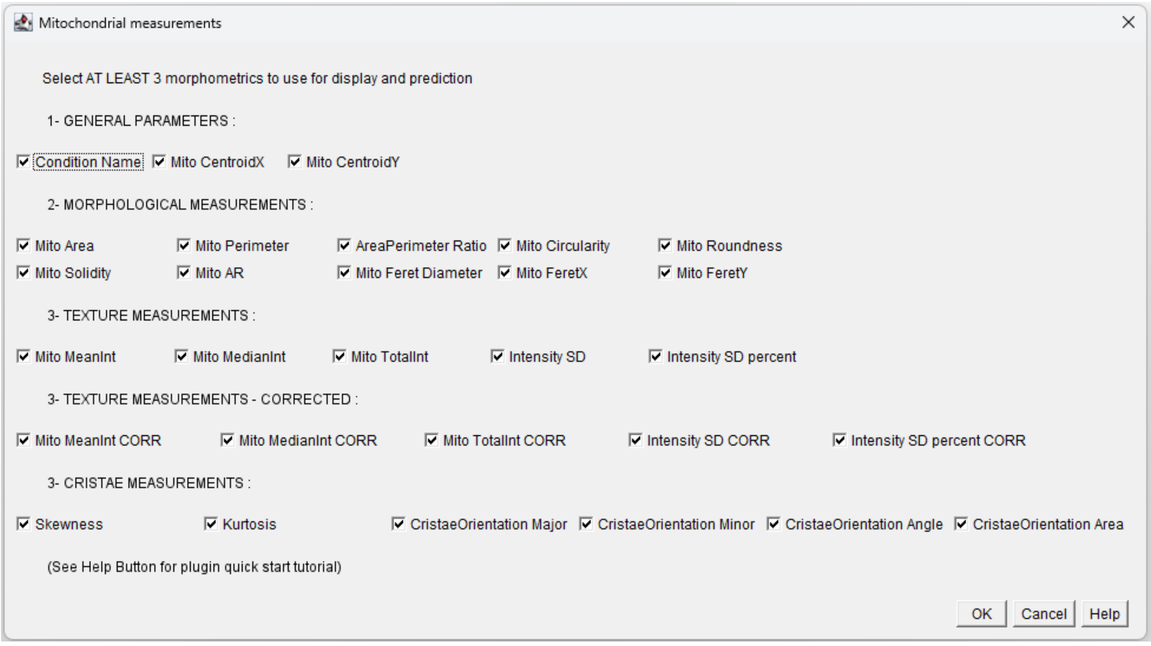
  <u>General parameters</u>: condition & images name, mitochondria labels and spatial position

  <u>Morphological measurements</u>: mitochondria shape measurements

  <u>Texture measurements</u>: ultrastructure measurements, based on distribution of normalized gray values in mitochondria

  <u>Texture measurements - corrected</u>: ultrastructure measurements, based on distribution of corrected gray values calculated in mitochondria. Corrected gray values are calculated from normalized gray values, by filtering noise with a high frequency filter (FFT)

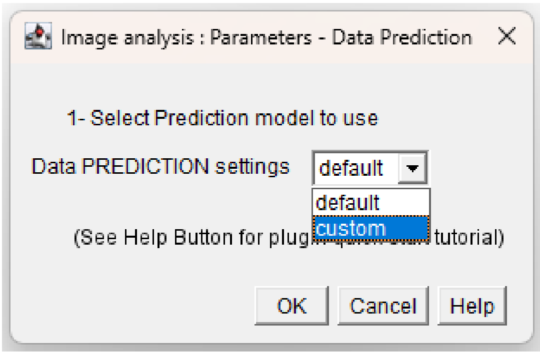
  <u>Cristae measurements</u>: measurements of cristae's organization and orientation



(see here for a detail description of mitometrics)

- D/ Selecting machine learning models for prediction

We developed a machine learning (ML) module with predictive analytic tools in order to determine how any given experimental condition would influence mitochondrial morphology and ultrastructure. In the next window, you have to select machine learning model to use for the predictive analysis.



- **Default option**: use all ML models for data prediction

- **Custom option:** manual selection of ML models for data prediction (select at least 1 model).

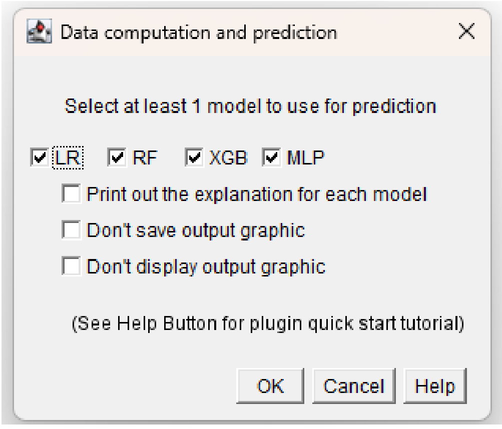
  Print out explanation: generates summary plots for each selected ML model (see here for description)

  Others output settings: don't save output graphic

  Graphs and distributions are displayed during the analysis, but not saved in the output folder

  Others output settings: don't display output graphic

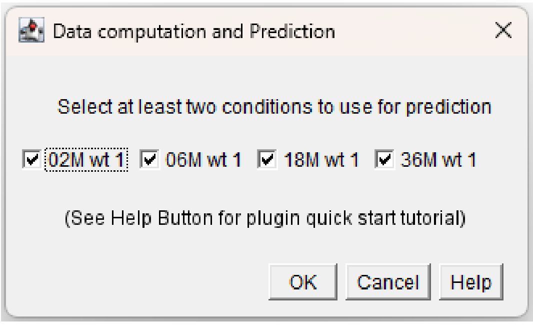
  Graphs and distributions are saved in the output folder, but not displayed during the analysis



(see [here](#) for a detail description of ML models)

- **E/ Selecting conditions for data computation & prediction**

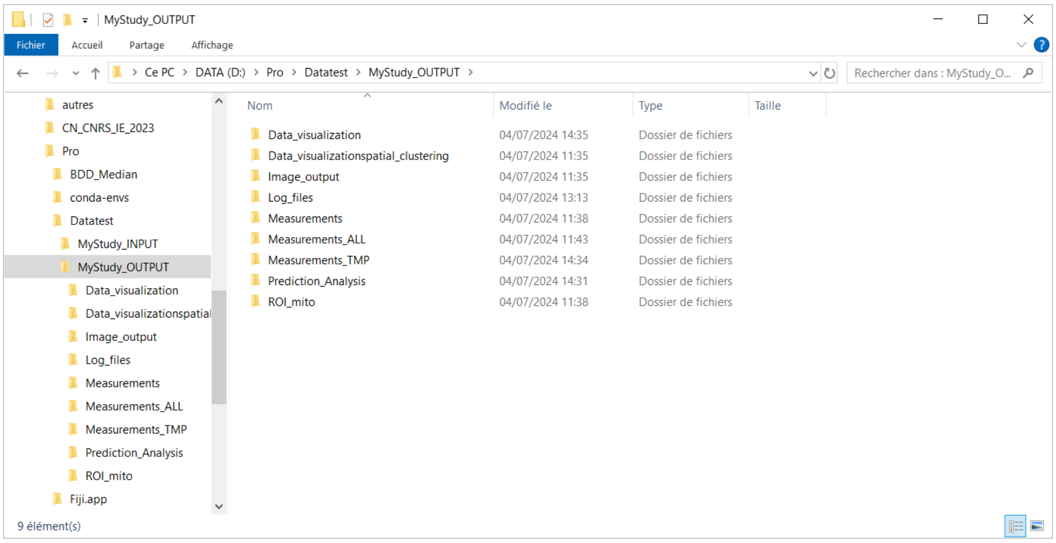
Select conditions to use for data prediction (switch on or switch off each condition). You must select at least two conditions (see below). Then data computation/prediction is starting; you will be able to follow the processing on the "LOG" Window.



Once data prediction is done, your mitometrix analysis is done. If needed, start again the EMitometrix application to restart one or more steps.

EMITO-METRIX OUTPUT DESCRIPTION

In this section, we describe folders, images, measurement files and data distributions saved in the output folders.

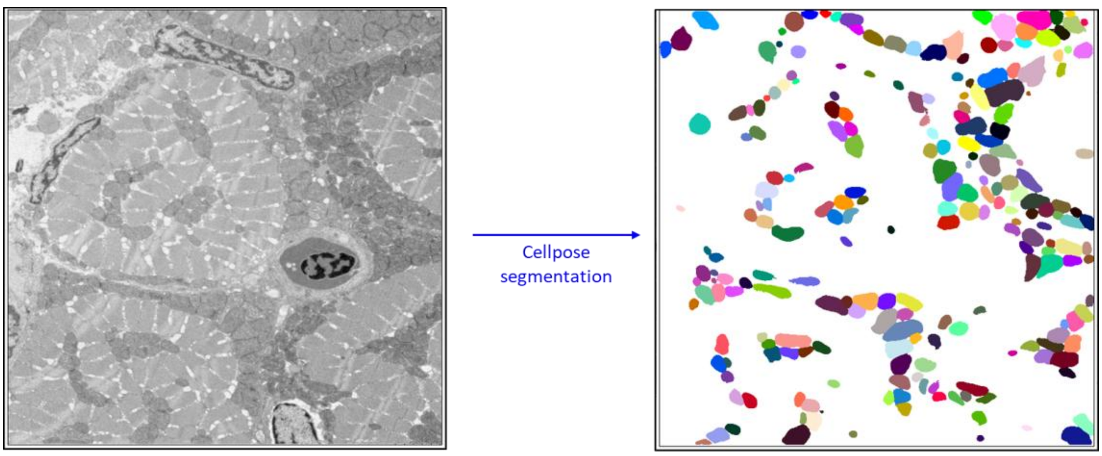


- ***Image_output*** **folder**

- *processed_images*: normalized & cropped EM images

- *cellpose_mask*: label maps containing segmented mitochondria

One value (i.e. one region of interest ROI) per mitochondria detected

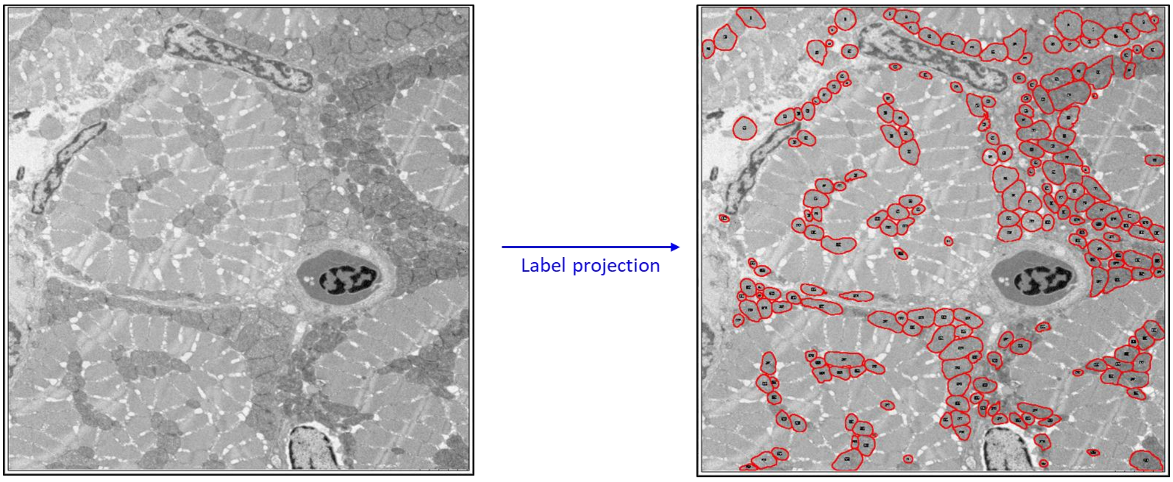


Pre-processed TEM image

Label map of mitochondria

- *control_quality_mask*: projection of mitochondria segmentation and normalized EM image

Used for the quality control of mitochondria detection



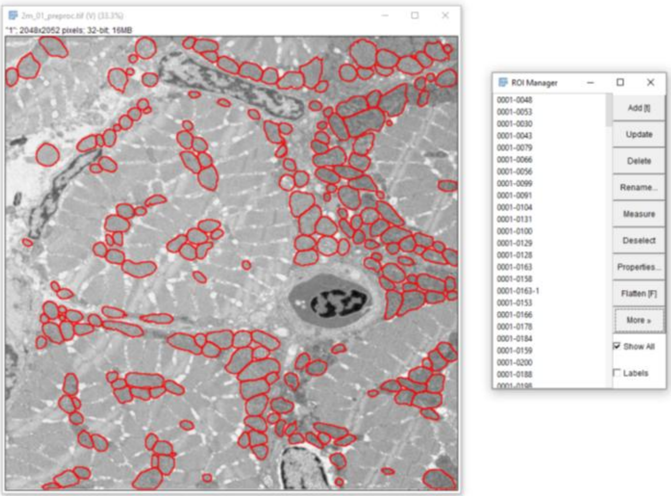
Pre-processed TEM image

Mitochondria label projection

- ***<u>ROI_mito</u>*** **<u>folder</u>**

Label regions of mitochondria detected with Cellpose or custom models. These regions of interest (ROI) files are opened and displayed using the *Fiji-ROI manager tool*

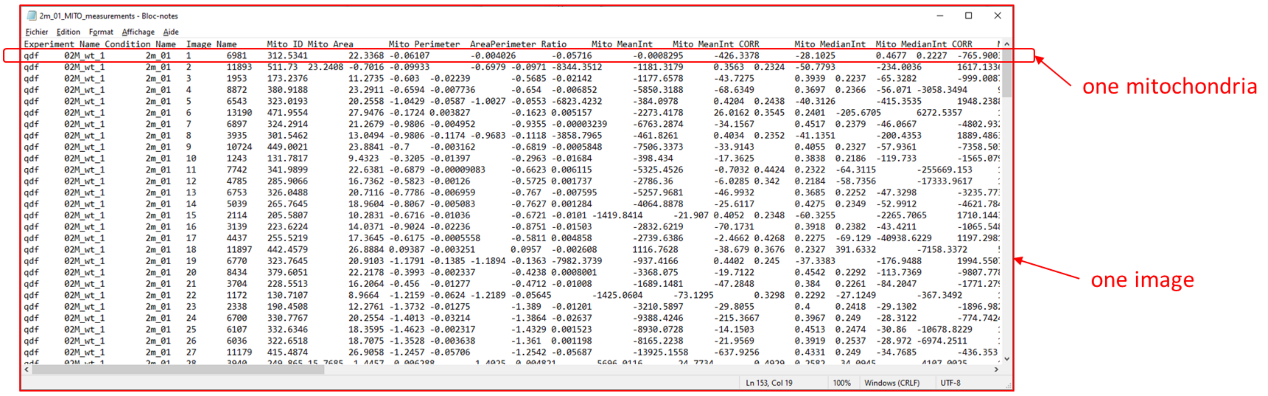
Drag and drop your EM image in Fiji, then open the ROI file using Analyze/Tools/ROI manager



- ***Measurements* folder**

Morphometrics files containing morphological and texture measurements (see here for a detailed description of mitochondrial measurements).

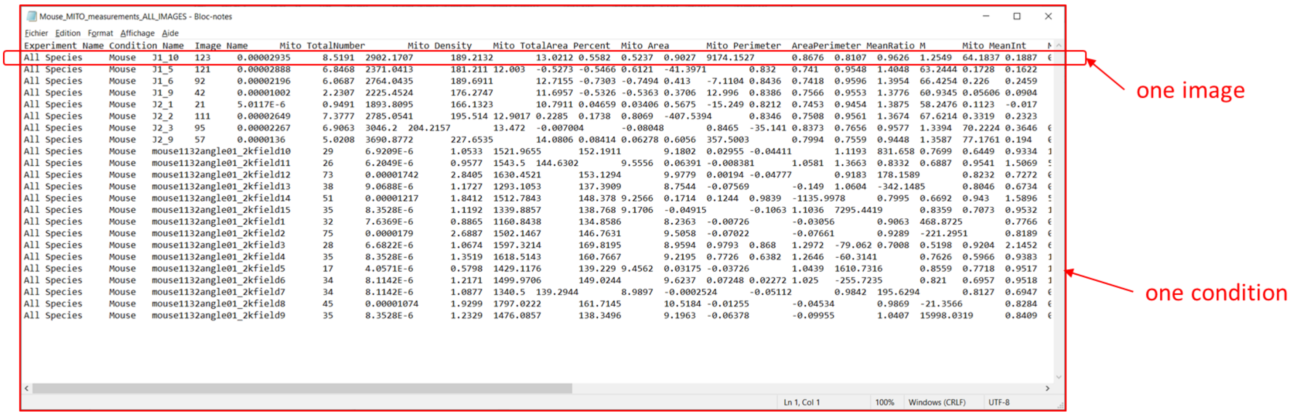
One text file per image and one line per mitochondria detected



- ***Measurements_ALL* folder**

Contains average morphological and texture measurements for each image (see here for a detailed description of mitochondrial measurements)

One text file per condition, and one line per image



- ***Measurements_TMP* folder**

Temporary folder used during application execution

- ***Data_visualization* folder**

Folder containing graphs and distributions for data visualization

__density.png__: density distribution of mitometrics for each condition

__histogram.png__: histogram distribution of mitometrics for each condition

__MinMaxScaler_radar_plot.png__: radar plot of MinMax rescaled mitometrics for each condition

__StandardScaler_radar_plot.png__: radar plot of Standard rescaled mitometrics for each condition

__PCA_Condition_Name.png__: PCA distribution of mitochondria according to the conditions

__UMAP_Condition_Name.png__: UMAP projection of mitochondria according to the conditions

__violin.png__: violin distribution of mitometrics for each condition

- **Data_visualizationspatial_clustering folder**

Folder containing spatial clustering of mitochondria detected for each image (*spatial_clustering* file), as well as density graph (*density* file) and mean morphometrics (*clusters* file) computed for each cluster

- ***Prediction_analysis* folder**

Folder containing SHAP values (*beeswarm* file) and confusion matrix (*confusion_matrix* file) for each machine learning algorithms used.

- **Log_files folder**

Folder containing application settings *(Users_general_parameters.txt)* and segmentation settings *(Users_Mitochondria_size.txt)*.

MITOCHONDRIAL MEASUREMENTS DESCRIPTION

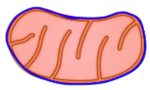
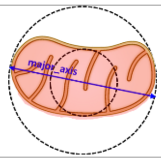
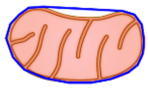
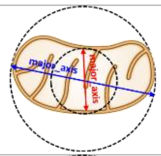
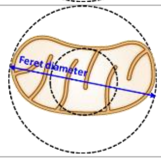
In this section, we describe mitochondrial measurements generated during the EMitoMetrix analysis

- **General parameters**

  Condition & images name, mitochondria labels and spatial position (Centroid X&Y)

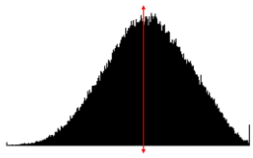
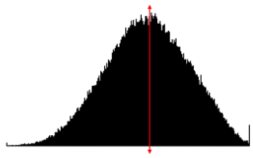
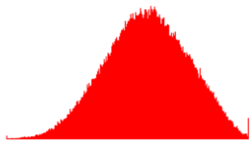
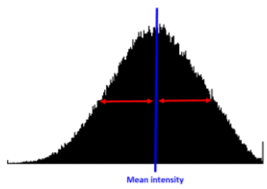
- **Morphological measurement**

  Mitochondria shape measurements

METRIC NAME	DEFINITION	Illustration
Mito_Area	Area of the mitochondria	
Mito_Perimeter	The length of the outside boundary of the mitochondria.	
AreaPerimeter_Ratio	Ratio between Perimeter and Area	
Mito_CentroidX and Mito_CentroidY	X and Y coordinates of the center point of the mitochondria.	
Mito_Circularity	Shape of the mitochondria defined as $(4\pi * area/perimeter^2)$. A value of 1.0 indicates a perfect circle. As the value approaches 0.0, it indicates an increasingly elongated shape	
Mito_Roundness	Shape of the mitochondria defined as $(4 * area/(\pi * major\_axis^2))$, or the inverse of the aspect ratio.	
Mito_Solidity	Ratio between area and convex area	
Mito_AR	Aspect Ratio : Ratio between major axis and minor axis	
Mito_Feret_Diameter	The longest distance between any two points along the mitochondria boundary, also known as maximum caliper	
Mito_FeretX and Mito_FeretY	Starting coordinates of the Feret's diameter	

- **Texture measurements**

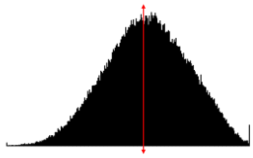
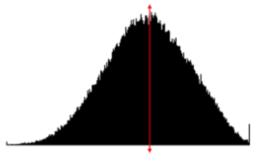
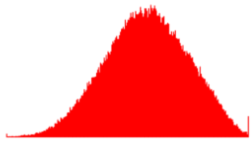
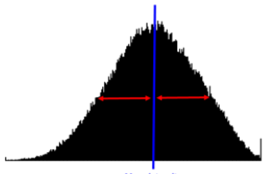
Ultrastructure measurements, based on distribution of normalized gray values in mitochondria

METRIC NAME	DEFINITION	Illustration
Mito_MeanInt	Average intensity calculated from mitochondria's normalized gray values	
Mito_MedianInt	Median intensity calculated from mitochondria's normalized gray values.	
Mito_TotalInt	Sum of the mitochondria's normalized gray values	
Intensity_SD	Standard deviation of the mitochondria's normalized gray values used to generate the mean intensity. Measure of crista's density within the mitochondria	
Intensity_SD_percent	Ratio between Intensity_SD and MeanInt. Measure of crista's density within the mitochondria	

- <u>**Texture measurements – corrected**</u>

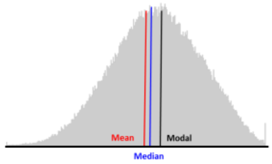
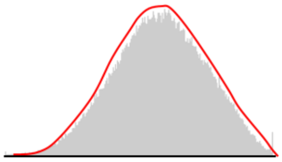
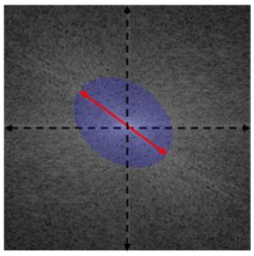
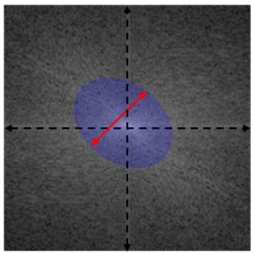
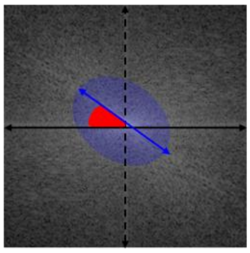
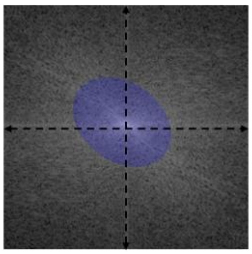
Ultrastructure measurements, based on distribution of corrected gray values calculated in mitochondria.

Corrected gray values are calculated from normalized gray values, by filtering noise with a high frequency filter (Fast Fourier Transformation).

METRIC NAME	DEFINITION	Illustration
Mito_MeanInt_CORR	Average intensity calculated from mitochondria's gray values after High frequency filtering (FFT noise correction)	
Mito_MedianInt_CORR	Median intensity calculated from mitochondria's gray values after High frequency filtering (FFT noise correction)	
Mito_TotalInt_CORR	Sum of the mitochondria's gray values, after High frequency filtering (FFT noise correction)	
Intensity_SD_CORR	Standard deviation of the mitochondria's gray values used to generate the mean, after High frequency filtering (FFT noise correction). Measure of crista's density within the mitochondria	
Intensity_SD_percent_CORR	Ratio between Intensity_SD_CORR and MeanInt_CORR. Measure of crista's density within the mitochondria	

- **Crista measurements**

Measurements of cristae's organization and orientation

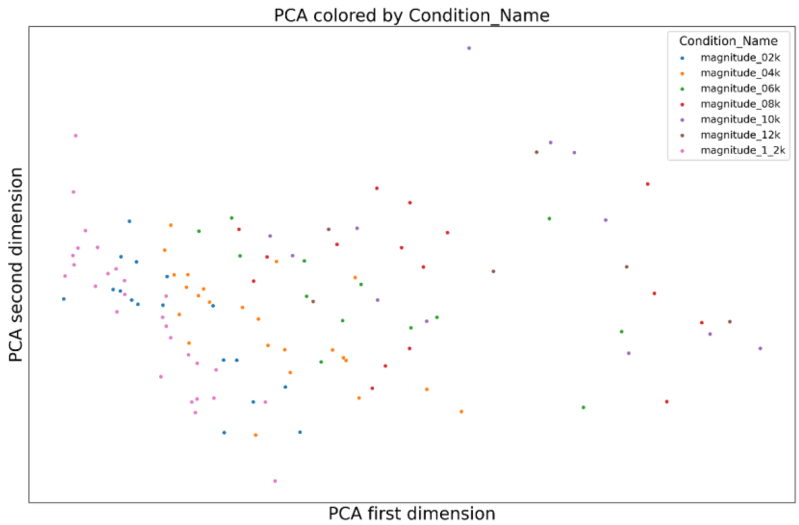
METRIC NAME	DEFINITION	Illustration
Skewness	Third order moment about the mean. Measure of the asymmetry of the mitochondria's normalized gray values about the mean intensity	
Kurtosis	The fourth order moment about the mean. Measure of the "tailedness" of the mitochondria's normalized gray values about the mean intensity.	
CristaOrientation_Major	Primary axis of the best fitting ellipse calculated from the frequency spectrum of the mitochondria's normalized gray values. Measure of the Crista's orientation, alignment and number within the mitochondria.	
CristaOrientation_Minor	Secondary axis of the best fitting ellipse calculated from the frequency spectrum of the mitochondria's normalized gray values. Measure of the Crista's orientation, alignment and number within the mitochondria.	
CristaOrientation_Angle	Angle (between the primary axis and a **line parallel to the x-axis** of the image) of the best fitting ellipse calculated from the frequency spectrum of the mitochondria's normalized gray values. Measure of the Crista's orientation, alignment and number within the mitochondria.	
CristaOrientation_Area	Area of the best fitting ellipse calculated from the frequency spectrum of the mitochondria's normalized gray values. Measure of the Crista's orientation, alignment and number within the mitochondria.	

GRAPHS & DATA DISTRIBUTION DESCRIPTION

In this section, we describe graphs and data distributions generated during the EMitoMetrix analysis

- **PCA distribution**

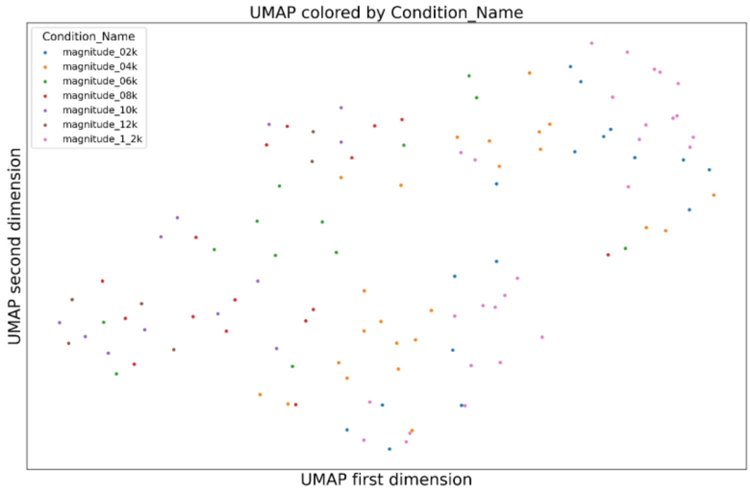
The principal component analysis (PCA) is a linear dimensionality reduction technique that consists in transforming data onto a new coordinate system such that dimensions are orthogonal and capture the most variation in the data. We keep only the first two dimensions to be plotted.



PCA distribution of mitochondria (one point per mitochondria), according to the magnitude used for EM acquisitions.

- **UMAP distribution**

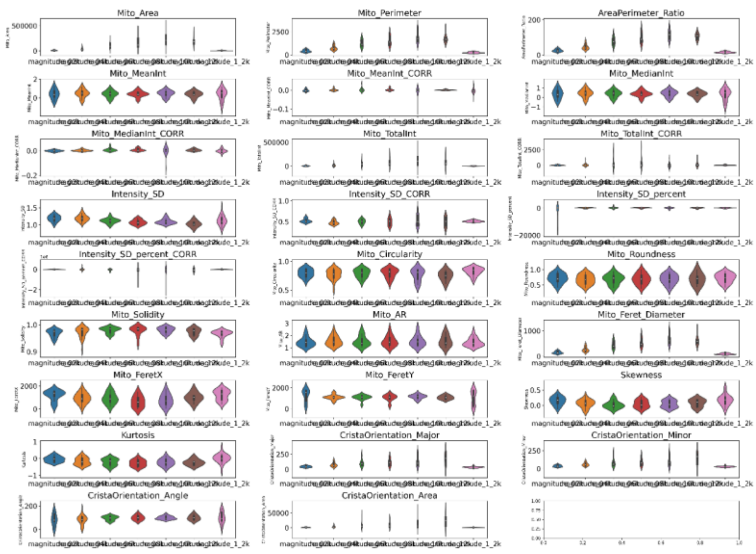
The Uniform Manifold Approximation for Projection (UMAP) is a non-linear dimensionality technique based on Rimannian manifold. It connects each point to its nearest neighbors in high dimension, and project the manifold in low dimension to be plotted. Its properties ensure that closest on the projection are close in high dimension, and points that are far from each other on the projection are far from each other in high dimension.



UMAP projection of mitochondria (one point per mitochondria), according to the magnitude used for EM acquisitions.

- **Violin distribution**

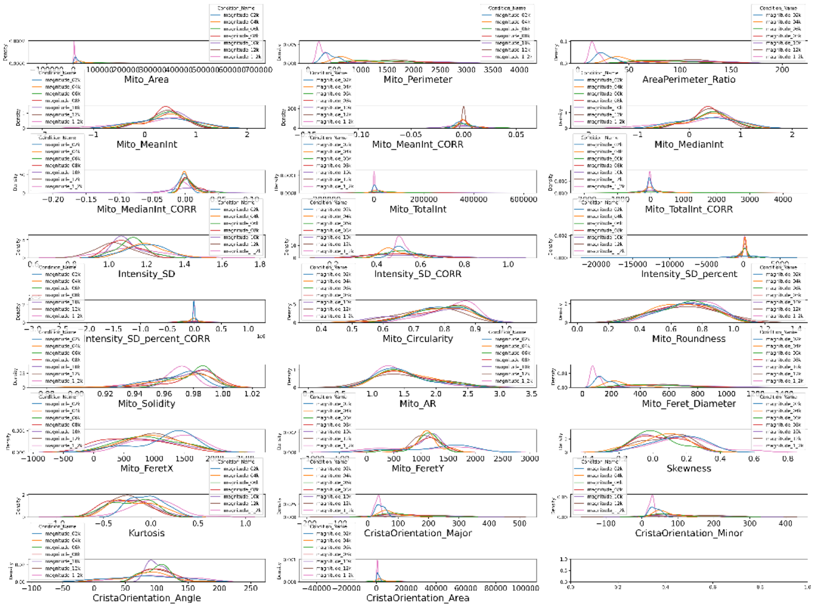
A violin plot is used to easily compare distributions between conditions. Each condition is plotted on a separate axis, but they are aligned with each other for comparison. The density is smoothed by a kernel density estimator.



Violin plots of mitochondria measurements (one plot per measurement), according to the magnitude used for EM acquisitions.

- **<u>Density distribution</u>**

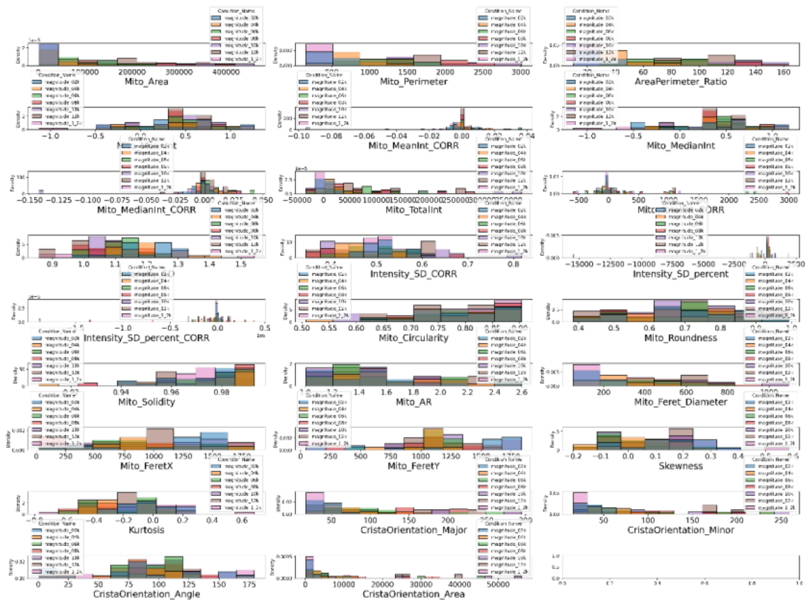
A density plot is used to represent the distribution of each condition, superimposed on each other. The line represent the kernel density estimator smoothed distribution of density.



Density distribution of mitochondria measurements (one distribution per measurement), according to the magnitude used for EM acquisitions.

- **<u>Histogram distribution</u>**

Similar to density plots, histograms are used to represent the density of the distribution of each condition, superimposed to each other. It depicts the distribution with more precision, as it is not smoothed by a kernel density estimator.

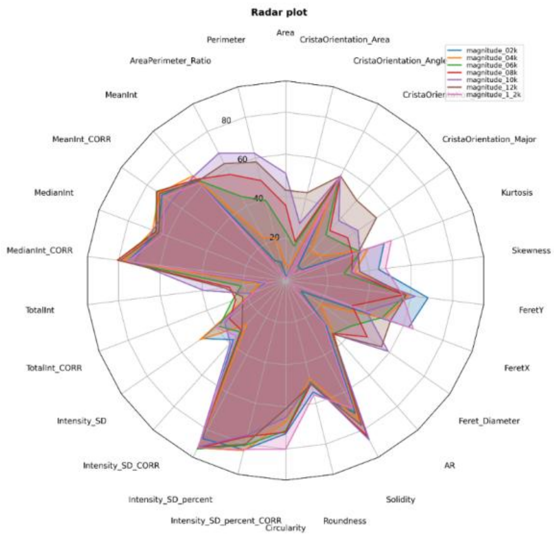


Histogram distribution of mitochondria measurements (one distribution per measurement), according to the magnitude used for EM acquisitions

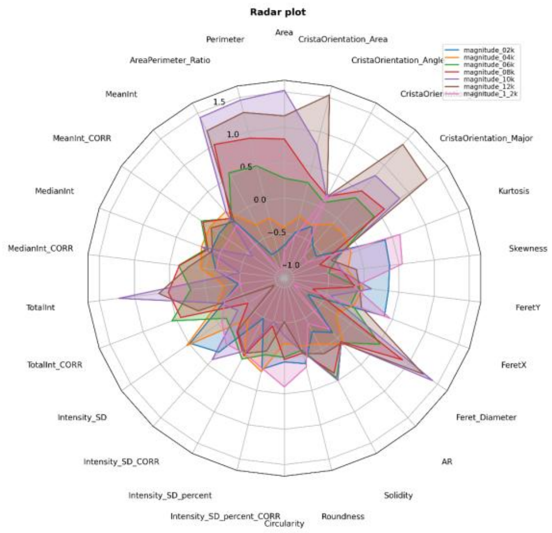
- **<u>Radar plot distribution</u>**

A radar plot (or radar chart) represent all the features of each condition on a single radial plot for comparison. To compare between features, two scaler have been used: a MinMax Scaler, rescaling all values between 0 and 1 for all features, and a Standard Scaler, substracting the mean value of each feature and dividing by its variance in order to have a pseudo-gaussian distribution centered on 0 with unit variance.

MinMax Scaler



Standard Scaler

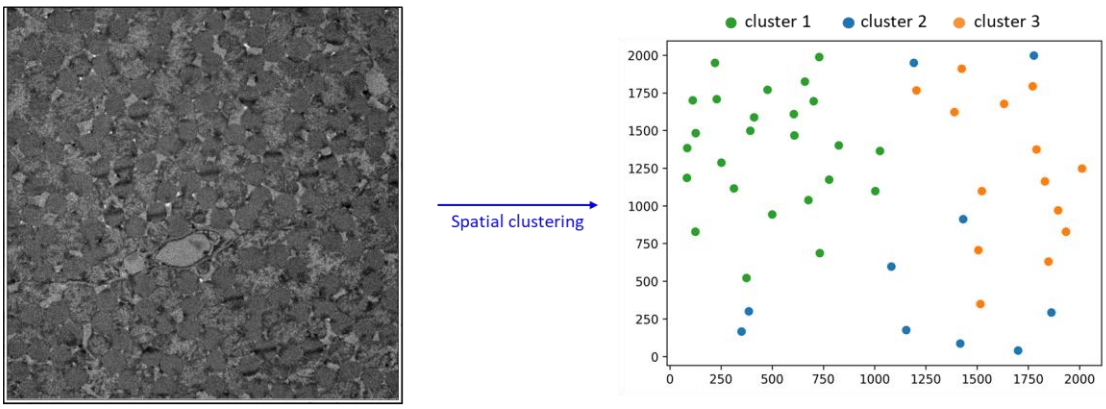


Radar plot distribution of mitochondria measurements according to the magnitude used for EM acquisitions.

- **<u>Spatial clustering distribution</u>**

The spatial clustering plot simply represent the physical position of the centroid of each mitochondria on a two-dimensional plot. A density-based clustering algorithm, HDBSCAN, is used to automatically capture groups of mitochondria, and each of them are averaged for each variable to be compiled in an output csv file.

!! Spatial clustering will increase calculation time !!



Pre-processed TEM image

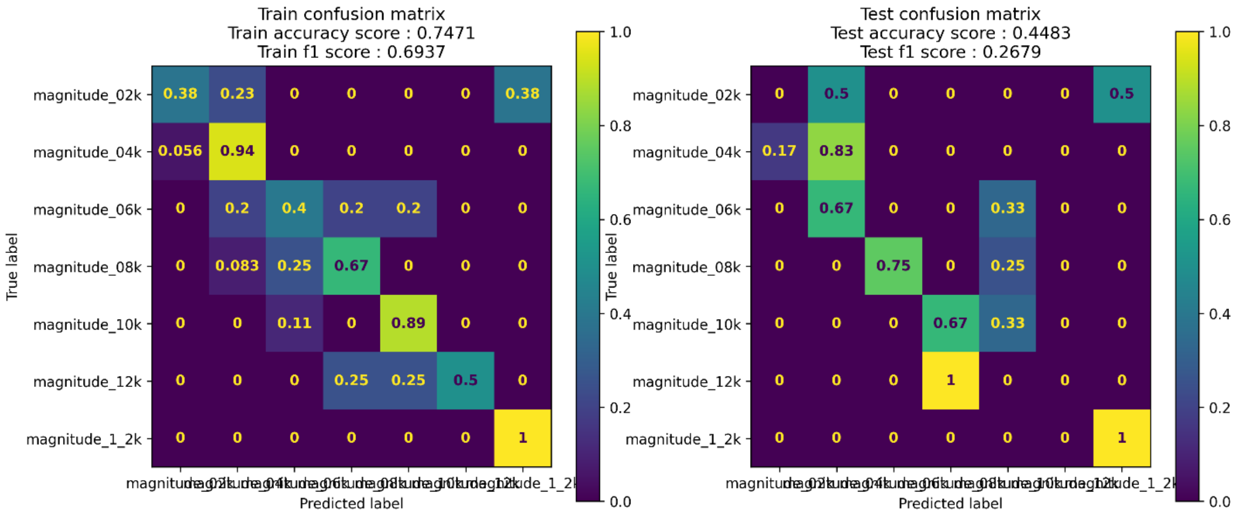
(X,Y) projection of Mitochondria's centroid

MACHINE LEARNING MODELS DESCRIPTION

In this section, we describe ML models used during the EMitoMetrix analysis, as well as confusion matrix and summary plots generated as output.

- **<u>Confusion Matrix</u>** (LR, RF, XGB, MLP)

A confusion matrix presents the predictive performances of a model (see below). The cell (i,j) represents the proportion of mitochondria of true class j that have been predicted of class i by the model. The diagonal (from top-left to bottom-right) represents the correct predictions. Train and test performances have been separated for overfitting assessment, and models performances in term of precision and f1-score are displayed above each matrix.



Four models are proposed for the computing of confusion matrix (see [here] for ML description):

<u>Linear Regression model (LR)</u>: tree-based algorithms

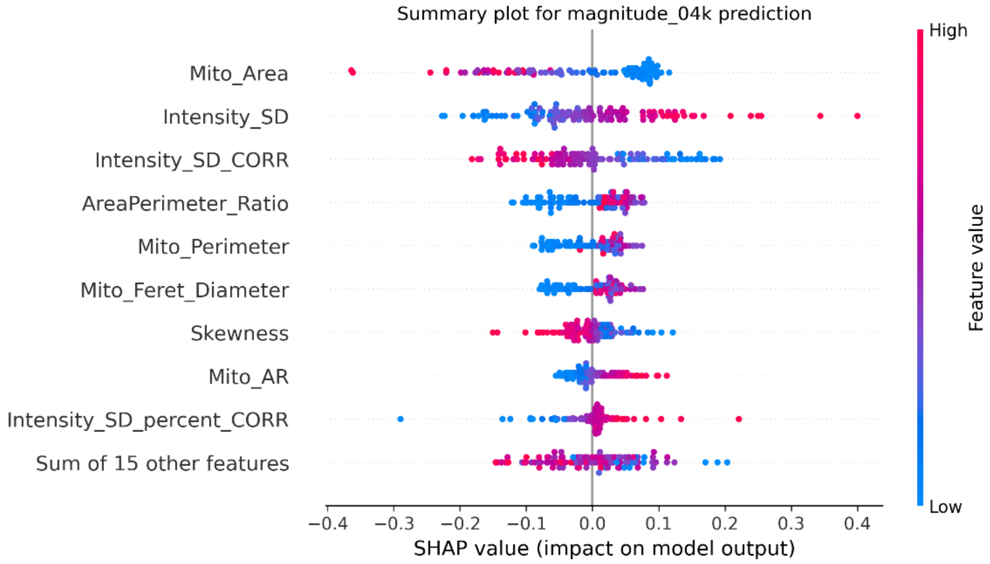
<u>Random Forrest model (RF)</u>: tree-based algorithms

<u>XGBoost model (XGB)</u>: tree-based algorithms

<u>MultiLayer Perceptron model (MLP)</u>: neural network

- **<u>Summary plots</u> (explanation option):**

A summary plot represents the explanation of a model, for a given class (see below). The feature are displayed from the most important (at the top) to the least important (at the bottom). Only the 14 most important features are displayed, and if there are more, they are summed together on a 15th axis. On each axis, each dot represent a mitochondria, its position on the x-axis represent the contribution of the associated feature to the prediction, and its color represent its relative feature value (red represents high values, while blue represents low values). For binary prediction, a single summary plot is displayed, because the summary plot for the second class is the exact opposite of the first one.



SHAP values generated with Linear Regression algorithms (LR), for the magnitude_04k condition

To compute summary plots, activate *print out the explanation function*